Parameter Estimation for Bilinear Time Series Models

Step-by-step derivations, algorithms, and numerical verification

Zaki Abiyu Aqilah

20 October 2026

Abstract

This document explains how to estimate the parameters of a bilinear time series model by *conditional maximum likelihood*, which turns out to be equivalent to minimising the sum of squared errors. Every formula is derived step by step: the log-likelihood function, the recursive derivatives of the errors, the gradient and the Hessian matrix, the Newton–Raphson (Gauss–Newton) iteration, standard errors, the AIC criterion, the invertibility condition, residual tests, and forecasting. The document is self-contained: every concept it uses is explained within it. At the end, the method is tested by recomputing three models for real data reported by Subba Rao & Gabr (1984); the recomputed results agree with the figures they report.

Contents

1 Introduction **2**

2 Model and notation **2**

2.1 The general bilinear model 2

2.2 The subset bilinear model 2

2.3 Conventions 3

2.4 Stationarity and invertibility (brief definitions) 3

3 Likelihood function and objective function **3**

3.1 The problem: the errors are not observed 3

3.2 Initial values 4

3.3 The joint density written as a product of conditional densities 4

3.4 The conditional density of each observation 4

3.5 The log-likelihood 5

3.6 Estimating σ^2 and reducing the problem to minimising Q 5

4 First derivatives of the errors **6**

4.1 Why the derivatives are recursive: a small example 6

4.2 General derivation 6

4.3 The three cases explicitly . . . 7
4.4 Initial values of the derivatives . . . 7
4.5 Extension to MA terms . . . 7

5 Gradient and Hessian matrix **9**
5.1 Basic concepts with one parameter . . . 9
5.2 Many parameters: the gradient vector and the Hessian matrix . . . 9
5.3 Example: linear regression and the normal equations . . . 10
5.4 The gradient for the bilinear model . . . 11
5.5 The Hessian for the bilinear model . . . 11
5.6 Numerical example . . . 12

6 Newton–Raphson and Gauss–Newton iteration **12**
6.1 Why iteration is needed . . . 12
6.2 Taylor series: approximating a function by a polynomial . . . 13
6.3 The Newton–Raphson method with one parameter . . . 13
6.4 Newton–Raphson with many parameters . . . 14
6.5 The Gauss–Newton method . . . 14
6.6 Step-size control . . . 14
6.7 Stopping rule . . . 15
6.8 Algorithm . . . 15

7 Starting values **16**
7.1 From the autoregressive model . . . 16
7.2 The repeated-residual method . . . 16
7.3 Staged starting values . . . 17
7.4 Multiple starting points (multi-start) . . . 17

8 Model selection **17**
8.1 Akaike's information criterion . . . 17
8.2 Stepwise selection procedure . . . 17

9 Standard errors **19**
9.1 Derivation . . . 19
9.2 Parameter significance test . . . 19

10 Invertibility **20**
10.1 Why it matters . . . 20
10.2 Propagation equation for the initial-value error . . . 20

10.3 The case of one bilinear term . . . 21
10.4 A practical diagnostic . . . 21

11 Residual tests **21**

11.1 The independence test W . . . 22
11.2 Normality test (Lomnicki) . . . 22
11.3 What to do if a test fails . . . 23

12 Forecasting **24**

12.1 One step ahead . . . 24
12.2 Two or more steps: derivation on an example . . . 24
12.3 General rule . . . 25
12.4 The Monte Carlo alternative . . . 25

13 A hand-computed example **26**

13.1 The error recursion . . . 26
13.2 Recursive derivatives . . . 26
13.3 One iteration step . . . 27

14 A manual Gauss–Newton iteration example **27**

14.1 Data and model . . . 27
14.2 Step 0: starting values . . . 28
14.3 First iteration ($s = 0$) . . . 28
14.4 Second iteration ($s = 1$) . . . 29
14.5 Further iterations and convergence . . . 30

15 Simulation study **31**

16 Verification with real data **32**

16.1 Model S: sunspots . . . 32
16.2 Model L1: log lynx . . . 32
16.3 Model L2: raw lynx . . . 32
16.4 Not yet verified . . . 33

17 Notes on figures printed in the book **33**

18 Numerical pitfalls and recommendations **33**

References **34**

1 Introduction

The bilinear model is an extension of the ARMA model that contains product terms between past values of the series and past errors. The main difficulty in estimating its parameters is that the errors e_t are *not observed*: they must be reconstructed from the data, and this reconstruction itself depends on the parameters being estimated. As a consequence, the function to be minimised is not quadratic, its derivatives must be computed recursively, and estimation must be carried out iteratively.

This document sets out all the computations required, so that it can serve as a basis for writing one's own estimation function. The method described is the one used by Subba Rao & Gabr (1984). The data and the reference results used for comparison also come from that book, but everything needed to understand the computations is written out again here.

Outline: Section 2 defines the model; Section 3 derives the likelihood function; Sections 4–6 derive the derivatives and the iteration; Sections 7–12 discuss starting values, model selection, standard errors, invertibility, residual tests, and forecasting; Sections 13–16 contain a hand-computed example, simulations, and verification with real data.

2 Model and notation

2.1 The general bilinear model

Definition 2.1 (Bilinear model BL(p, r, m, k))**.** A time series $\{X_t\}$ follows a bilinear model if

$$X_t + \sum_{j=1}^{p} a_j X_{t-j} \;=\; \alpha + \sum_{j=1}^{r} c_j e_{t-j} + \sum_{i=1}^{m}\sum_{j=1}^{k} b_{ij}\, X_{t-i}\, e_{t-j} + e_t, \tag{1}$$

where $\{e_t\}$ is a sequence of independent, identically distributed $N(0, \sigma^2)$ random variables.

The model has four parts: the autoregressive part (a_j), the constant (α), the moving-average part (c_j), and the bilinear part (b_{ij}). If all $b_{ij} = 0$, the model reduces to ARMA(p, r).

2.2 The subset bilinear model

In practice, often only a few of the coefficients are really needed. A *subset* model uses only selected lags.

Definition 2.2 (Subset bilinear model, SBL)**.** Let $\mathcal{K} = \{k_1, \ldots, k_\ell\}$ be the set of autoregressive lags and $\mathcal{B} = \{(r_1, s_1), \ldots, (r_m, s_m)\}$ the set of bilinear lag pairs. The subset bilinear model is

$$X_t + \sum_{k\in\mathcal{K}} a_k X_{t-k} \;=\; \alpha + \sum_{(i,j)\in\mathcal{B}} b_{ij}\, X_{t-i}\, e_{t-j} + e_t. \tag{2}$$

Its parameter vector is

$$\boldsymbol{\theta} = (a_{k_1}, \ldots, a_{k_\ell},\; b_{r_1 s_1}, \ldots, b_{r_m s_m},\; \alpha)',$$

of dimension $q = \ell + m + 1$. In addition to $\boldsymbol{\theta}$, there is the variance parameter σ^2.

Most of this document uses the subset model (2), because the full model (1) without MA terms is a special case of it (take $\mathcal{K} = \{1, \ldots, p\}$ and let $\mathcal{B}$ contain all pairs). The extension to MA terms is treated separately in Section 4.5.

2.3 Conventions

Conventions used throughout the document

1. Autoregressive coefficients are written on the **left-hand side**. The model "$X_t = 0.4X_{t-1} + \ldots$" means $a_1 = -0.4$.
2. In b_{ij}, the first index is the lag of X and the second the lag of e: b_{ij} is the coefficient of $X_{t-i}\, e_{t-j}$. So that the term $X_{t-i}e_{t-j}$ contains only the past, always $i \geq 1$ and $j \geq 1$.
3. The model is fitted to the **original data** (not mean-corrected) with a constant α.
4. γ denotes the largest lag used, on either X or e.

Other literature sometimes uses a different sign convention, for example $x_t = c + \sum \phi_i x_{t-i} + \sum\sum \beta_{ij} x_{t-i} e_{t-j} + e_t$. The relation is: $\phi_i = -a_i$, $c = \alpha$, $\beta_{ij} = b_{ij}$.

2.4 Stationarity and invertibility (brief definitions)

Stationarity. The model must generate a stationary series (one whose statistical properties do not change over time). For the special case BL$(p, 0, p, 1)$, i.e. model (1) with $r = 0$, $m = p$, $k = 1$, define the $p \times p$ matrices

$$\mathbf{A} = \begin{pmatrix} -a_1 & -a_2 & \cdots & -a_{p-1} & -a_p \\ 1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & 1 & 0 \end{pmatrix}, \qquad \mathbf{B} = \begin{pmatrix} b_{11} & b_{21} & \cdots & b_{p1} \\ 0 & 0 & \cdots & 0 \\ \vdots & & & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}.$$

Subba Rao (1981) showed that this model is second-order stationary if

$$\rho(\mathbf{A}) < 1 \quad \text{and} \quad \rho(\mathbf{A} \otimes \mathbf{A} + \sigma^2\, \mathbf{B} \otimes \mathbf{B}) < 1,$$

where $\rho(\cdot)$ is the spectral radius (the largest absolute value of the eigenvalues) and $\otimes$ is the Kronecker product ($\mathbf{A} \otimes \mathbf{B}$ is the block matrix whose (i, j)-th block is $a_{ij}\mathbf{B}$). For more general subset models the stationarity condition is more complicated and is not discussed here. This condition is used to *check* the estimates, not to compute them.

Invertibility. The model is called invertible if the errors e_t can be reconstructed from past data with a reconstruction error that becomes smaller over time. This property is crucial for estimation and is discussed in detail in Section 10.

3 Likelihood function and objective function

3.1 The problem: the errors are not observed

The available data are $X_1, \ldots, X_N$. The errors e_t are not observed. However, equation (2) can be rearranged so that e_t is on the left-hand side:

$$\boxed{e_t = X_t + \sum_{k \in \mathcal{K}} a_k X_{t-k} - \alpha - \sum_{(i,j) \in \mathcal{B}} b_{ij}\, X_{t-i}\, e_{t-j}.} \tag{3}$$

The right-hand side contains e_{t-j} with $j \geq 1$, i.e. *earlier* errors. So if the earlier errors are known, e_t can be computed. This gives a way to compute the errors one after another (recursively).

3.2 Initial values

Recursion (3) needs a starting point. The largest lag is γ, so computing e_t requires $X_{t-1}, \ldots, X_{t-\gamma}$ and $e_{t-1}, \ldots, e_{t-\gamma}$. We:

1. treat $X_1, \ldots, X_\gamma$ as fixed initial values (we condition on them);
2. set
$$e_1 = e_2 = \cdots = e_\gamma = 0, \tag{4}$$
because their values are unknown and their expectation is zero;
3. compute $e_{\gamma+1}, e_{\gamma+2}, \ldots, e_N$ in turn with (3).

The number of computed errors is
$$n = N - \gamma.$$
Setting (4) is an approximation. It is a good one only if the effect of the wrong initial values dies out over time, i.e. if the model is invertible (Section 10).

Because the computed e_t depends on the parameter values, we write it as $e_t(\boldsymbol{\theta})$.

3.3 The joint density written as a product of conditional densities

Let $\mathcal{F}_{t-1}$ denote the information up to time $t-1$, i.e. $X_1, \ldots, X_{t-1}$. By the multiplication rule of probability, the joint density of $X_{\gamma+1}, \ldots, X_N$ (conditional on $X_1, \ldots, X_\gamma$) can be written
$$f(X_{\gamma+1}, \ldots, X_N \mid X_1, \ldots, X_\gamma) = \prod_{t=\gamma+1}^{N} f(X_t \mid \mathcal{F}_{t-1}). \tag{5}$$
This is just repeated application of $f(u, v) = f(v \mid u)\, f(u)$: the joint density equals the density of $X_{\gamma+1}$ given the past, times the density of $X_{\gamma+2}$ given its past, and so on.

3.4 The conditional density of each observation

Write (2) as
$$X_t = \mu_t + e_t, \qquad \mu_t = -\sum_{k \in \mathcal{K}} a_k X_{t-k} + \alpha + \sum_{(i,j) \in \mathcal{B}} b_{ij} X_{t-i} e_{t-j}.$$
All components of μ_t are known once $\mathcal{F}_{t-1}$ is known: X_{t-k} and X_{t-i} are past data, and e_{t-j} is computed from past data by recursion (3). So, *given* $\mathcal{F}_{t-1}$, μ_t is a constant. Meanwhile e_t is independent of the past and distributed $N(0, \sigma^2)$. Since X_t is the constant μ_t plus the normal variable e_t,
$$X_t \mid \mathcal{F}_{t-1} \;\sim\; N(\mu_t,\, \sigma^2),$$
so that
$$f(X_t \mid \mathcal{F}_{t-1}) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{(X_t - \mu_t)^2}{2\sigma^2} \Big) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(-\frac{e_t(\boldsymbol{\theta})^2}{2\sigma^2} \Big),$$
because $X_t - \mu_t = e_t(\boldsymbol{\theta})$ by (3).

Note

Another way to see this step is as a *change of variables*. The mapping from $(e_{\gamma+1}, \dots, e_N)$ to $(X_{\gamma+1}, \dots, X_N)$ is triangular: X_t depends only on e_t and earlier values, with $\partial X_t / \partial e_t = 1$. The Jacobian determinant of a triangular matrix is the product of its diagonal, which is 1. Hence the density of X equals the density of e evaluated at $e_t(\boldsymbol{\theta})$, with no multiplying factor.

3.5 The log-likelihood

Substitute the conditional densities into (5). The conditional likelihood function is

$$L(\boldsymbol{\theta}, \sigma^2) = \prod_{t=\gamma+1}^{N} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(- \frac{e_t(\boldsymbol{\theta})^2}{2\sigma^2} \Big).$$

Take logarithms. The logarithm of a product is the sum of the logarithms, and for each factor

$$\log \Big[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big(- \frac{e_t^2}{2\sigma^2} \Big) \Big] = -\tfrac{1}{2} \log(2\pi) - \tfrac{1}{2} \log \sigma^2 - \frac{e_t^2}{2\sigma^2}.$$

Summing the n factors gives

$$\boxed{\log L(\boldsymbol{\theta}, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{Q(\boldsymbol{\theta})}{2\sigma^2}, \qquad Q(\boldsymbol{\theta}) = \sum_{t=\gamma+1}^{N} e_t(\boldsymbol{\theta})^2.} \tag{6}$$

3.6 Estimating σ^2 and reducing the problem to minimising Q

For fixed $\boldsymbol{\theta}$, maximise (6) with respect to σ^2. The derivative is

$$\frac{\partial \log L}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{Q(\boldsymbol{\theta})}{2\sigma^4}.$$

Setting it to zero gives $-n\sigma^2 + Q(\boldsymbol{\theta}) = 0$, so

$$\hat{\sigma}^2(\boldsymbol{\theta}) = \frac{Q(\boldsymbol{\theta})}{n}. \tag{7}$$

Substitute back into (6). The last term becomes $Q/(2Q/n) = n/2$, so

$$\log L(\boldsymbol{\theta}, \hat{\sigma}^2(\boldsymbol{\theta})) = -\frac{n}{2}\big(1 + \log 2\pi\big) - \frac{n}{2} \log \frac{Q(\boldsymbol{\theta})}{n}. \tag{8}$$

Function (8) is called the *profile log-likelihood*. The log function is monotonically increasing, so $-\frac{n}{2}\log(\cdot)$ is monotonically decreasing. Hence maximising (8) with respect to $\boldsymbol{\theta}$ is the same as minimising $Q(\boldsymbol{\theta})$:

$$\boxed{\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} Q(\boldsymbol{\theta}), \qquad \hat{\sigma}^2 = Q(\hat{\boldsymbol{\theta}})/n.}$$

This method is called *conditional least squares* (CLS); under the normality assumption it is identical to *conditional maximum likelihood estimation* (conditional MLE).

Note

The value $n = N - \gamma$ must be the *same* for all models being compared, because criteria such as AIC (Section 8) are only meaningful when computed on the same observations. Therefore γ is fixed once at the start as the largest lag that could be used by any of the models to be compared.

4 First derivatives of the errors

Minimising $Q(\boldsymbol{\theta})$ requires its derivatives, and the derivatives of Q require the derivatives of each $e_t(\boldsymbol{\theta})$.

4.1 Why the derivatives are recursive: a small example

Take a model with one AR term and one bilinear term, $X_t + aX_{t-1} = \alpha + bX_{t-1}e_{t-1} + e_t$, with $\gamma = 1$. The recursion is $e_t = X_t + aX_{t-1} - \alpha - bX_{t-1}e_{t-1}$, with $e_1 = 0$. Then

$$\begin{aligned} e_2 &= X_2 + aX_1 - \alpha - bX_1 \cdot 0 = X_2 + aX_1 - \alpha, \\ e_3 &= X_3 + aX_2 - \alpha - bX_2\, e_2 = X_3 + aX_2 - \alpha - bX_2\,(X_2 + aX_1 - \alpha). \end{aligned}$$

We see that e_3 depends on b *twice*: directly through $-bX_2(\cdot)$, and indirectly through e_2, which also depends on a and α. Differentiating with respect to a:

$$\frac{\partial e_3}{\partial a} = X_2 - bX_2 \frac{\partial e_2}{\partial a} = X_2 - bX_2X_1.$$

The term $-bX_2X_1$ comes from the dependence of e_2 on a. If this term is ignored, the derivative is wrong. The larger t is, the longer this chain of dependence becomes, because e_t depends on e_{t-1}, which depends on e_{t-2}, and so on. Fortunately, the chain can be handled recursively, as derived below.

4.2 General derivation

Denote the derivative of the error with respect to the u-th parameter by

$$\psi_{t,u} = \frac{\partial e_t(\boldsymbol{\theta})}{\partial \theta_u}, \qquad u = 1, \ldots, q.$$

Differentiate both sides of (3) with respect to θ_u, term by term:

1. $\dfrac{\partial X_t}{\partial \theta_u} = 0$, because X_t is data, not a function of the parameters.
2. For the AR terms, X_{t-k} is also data, so $\dfrac{\partial}{\partial \theta_u}\big(a_k X_{t-k}\big) = X_{t-k}\dfrac{\partial a_k}{\partial \theta_u}$, which equals X_{t-k} if $\theta_u = a_k$ and 0 otherwise.
3. $\dfrac{\partial \alpha}{\partial \theta_u}$ equals 1 if $\theta_u = \alpha$ and 0 otherwise.
4. The bilinear term $b_{ij}X_{t-i}e_{t-j}(\boldsymbol{\theta})$ is a product of two quantities that depend on $\boldsymbol{\theta}$, namely b_{ij} and $e_{t-j}(\boldsymbol{\theta})$ (X_{t-i} is data). By the product rule $(fg)' = f'g + fg'$:

$$\frac{\partial}{\partial \theta_u}\big(b_{ij}X_{t-i}e_{t-j}\big) = \frac{\partial b_{ij}}{\partial \theta_u}\, X_{t-i}e_{t-j} + b_{ij}X_{t-i}\,\frac{\partial e_{t-j}}{\partial \theta_u} = \frac{\partial b_{ij}}{\partial \theta_u}\, X_{t-i}e_{t-j} + b_{ij}X_{t-i}\,\psi_{t-j,u}.$$

Combine the four parts with their signs from (3):

$$\psi_{t,u} = \sum_{k \in \mathcal{K}} X_{t-k}\frac{\partial a_k}{\partial \theta_u} - \frac{\partial \alpha}{\partial \theta_u} - \sum_{(i,j)\in\mathcal{B}} \frac{\partial b_{ij}}{\partial \theta_u} X_{t-i}e_{t-j} - \sum_{(i,j)\in\mathcal{B}} b_{ij}X_{t-i}\,\psi_{t-j,u}.$$

Of the first three sums only one term remains (the term for the parameter θ_u itself); call it the *direct term* $d_{t,u}$. The last sum is the *recursive term*. Thus

$$\psi_{t,u} = d_{t,u} - \sum_{(i,j)\in\mathcal{B}} b_{ij}\, X_{t-i}\, \psi_{t-j,u}, \qquad d_{t,u} = \begin{cases} X_{t-k} & \text{if } \theta_u = a_k, \\ -X_{t-r}\, e_{t-s} & \text{if } \theta_u = b_{rs}, \\ -1 & \text{if } \theta_u = \alpha. \end{cases} \tag{9}$$

4.3 The three cases explicitly

Writing (9) out for each type of parameter gives:

$$\boxed{\begin{aligned} \frac{\partial e_t}{\partial a_k} &= X_{t-k} - \sum_{(i,j)\in\mathcal{B}} b_{ij} X_{t-i} \frac{\partial e_{t-j}}{\partial a_k}, \\ \frac{\partial e_t}{\partial b_{rs}} &= -X_{t-r} e_{t-s} - \sum_{(i,j)\in\mathcal{B}} b_{ij} X_{t-i} \frac{\partial e_{t-j}}{\partial b_{rs}}, \\ \frac{\partial e_t}{\partial \alpha} &= -1 - \sum_{(i,j)\in\mathcal{B}} b_{ij} X_{t-i} \frac{\partial e_{t-j}}{\partial \alpha}. \end{aligned}} \tag{10}$$

Note that the recursive term is the same for all three types of parameter; only the direct term differs.

4.4 Initial values of the derivatives

Since $e_t = 0$ for $t \le \gamma$ is a constant that does not depend on $\boldsymbol{\theta}$, its derivatives are also zero:

$$\psi_{t,u} = 0, \qquad t \le \gamma,\ u = 1, \ldots, q.$$

Thus, for each $t = \gamma + 1, \ldots, N$, we compute e_t by (3) and then $\psi_{t,u}$ by (10), using the e and ψ already computed at earlier times. Both can be computed in the same pass.

The correctness of these formulas can be checked numerically by finite differences, $\psi_{t,u} \approx \big[e_t(\boldsymbol{\theta} + h\mathbf{1}_u) - e_t(\boldsymbol{\theta} - h\mathbf{1}_u)\big]/(2h)$ for small h (Section 13).

4.5 Extension to MA terms

This section re-derives the error recursion and its derivatives when the model also contains a moving-average (MA) part.

Step 1: the model. Let $\mathcal{C}$ be the set of MA lags. The subset model with MA terms is

$$X_t + \sum_{k\in\mathcal{K}} a_k X_{t-k} = \alpha + \sum_{l\in\mathcal{C}} c_l\, e_{t-l} + \sum_{(i,j)\in\mathcal{B}} b_{ij} X_{t-i} e_{t-j} + e_t,$$

and the parameter vector is extended by the coefficients c_l, $l \in \mathcal{C}$.

Step 2: the error recursion. Move all terms except e_t to the left-hand side:

$$e_t = X_t + \sum_{k\in\mathcal{K}} a_k X_{t-k} - \alpha - \sum_{l\in\mathcal{C}} c_l\, e_{t-l} - \sum_{(i,j)\in\mathcal{B}} b_{ij} X_{t-i} e_{t-j}. \tag{11}$$

Past errors now appear in *two* places: in the MA terms and in the bilinear terms. The initial values are still $e_t = 0$ for $t \le \gamma$, where γ now also covers the largest MA lag.

Step 3: differentiate term by term. Differentiate (11) with respect to θ_u. Four types of term are treated as before (X_t is data; AR terms; the constant; bilinear terms with the product rule). The MA term $c_l\, e_{t-l}(\boldsymbol{\theta})$ is also a product of two quantities that depend on $\boldsymbol{\theta}$, namely c_l and $e_{t-l}(\boldsymbol{\theta})$, so by the product rule

$$\frac{\partial}{\partial \theta_u}\big(c_l\, e_{t-l}\big) = \frac{\partial c_l}{\partial \theta_u}\, e_{t-l} + c_l\, \frac{\partial e_{t-l}}{\partial \theta_u} = \frac{\partial c_l}{\partial \theta_u}\, e_{t-l} + c_l\, \psi_{t-l,u}.$$

The first term equals e_{t-l} only if $\theta_u = c_l$, and zero for the other parameters.

Step 4: combine. With the signs from (11),

$$\psi_{t,u} = \underbrace{\sum_k X_{t-k}\frac{\partial a_k}{\partial \theta_u} - \frac{\partial \alpha}{\partial \theta_u} - \sum_l \frac{\partial c_l}{\partial \theta_u} e_{t-l} - \sum_{(i,j)} \frac{\partial b_{ij}}{\partial \theta_u} X_{t-i} e_{t-j}}_{\text{direct term } d_{t,u}}$$
$$- \sum_{l\in\mathcal{C}} c_l\, \psi_{t-l,u} \; - \sum_{(i,j)\in\mathcal{B}} b_{ij} X_{t-i}\, \psi_{t-j,u}.$$

The direct term leaves only one term for each parameter, giving four formulas:

$$\boxed{\begin{aligned}
\frac{\partial e_t}{\partial a_k} &= X_{t-k} && - \textstyle\sum_l c_l \frac{\partial e_{t-l}}{\partial a_k} - \sum_{(i,j)} b_{ij} X_{t-i} \frac{\partial e_{t-j}}{\partial a_k},\\
\frac{\partial e_t}{\partial c_m} &= -e_{t-m} && - \textstyle\sum_l c_l \frac{\partial e_{t-l}}{\partial c_m} - \sum_{(i,j)} b_{ij} X_{t-i} \frac{\partial e_{t-j}}{\partial c_m},\\
\frac{\partial e_t}{\partial b_{rs}} &= -X_{t-r} e_{t-s} && - \textstyle\sum_l c_l \frac{\partial e_{t-l}}{\partial b_{rs}} - \sum_{(i,j)} b_{ij} X_{t-i} \frac{\partial e_{t-j}}{\partial b_{rs}},\\
\frac{\partial e_t}{\partial \alpha} &= -1 && - \textstyle\sum_l c_l \frac{\partial e_{t-l}}{\partial \alpha} - \sum_{(i,j)} b_{ij} X_{t-i} \frac{\partial e_{t-j}}{\partial \alpha}.
\end{aligned}} \tag{12}$$

Compared with (10), there are two changes: a new formula for c_m appears with direct term $-e_{t-m}$, and every formula gets an additional recursive term $-\sum_l c_l\, \partial e_{t-l}/\partial \theta_u$.

Example. Take one AR term, one MA term, and one bilinear term, all at lag 1:

$$X_t + aX_{t-1} = \alpha + c\, e_{t-1} + b\, X_{t-1} e_{t-1} + e_t.$$

The error recursion can be written compactly because the MA term and the bilinear term both contain e_{t-1}:

$$e_t = X_t + aX_{t-1} - \alpha - (c + bX_{t-1})\, e_{t-1}.$$

The derivatives follow (12), with the same recursive multiplier $(c + bX_{t-1})$ for all parameters:

$$\frac{\partial e_t}{\partial a} = X_{t-1} - (c + bX_{t-1})\frac{\partial e_{t-1}}{\partial a}, \qquad \frac{\partial e_t}{\partial c} = -e_{t-1} - (c + bX_{t-1})\frac{\partial e_{t-1}}{\partial c},$$

$$\frac{\partial e_t}{\partial b} = -X_{t-1}e_{t-1} - (c + bX_{t-1})\frac{\partial e_{t-1}}{\partial b}, \qquad \frac{\partial e_t}{\partial \alpha} = -1 - (c + bX_{t-1})\frac{\partial e_{t-1}}{\partial \alpha}.$$

With data $X = (1.2,\ 0.8,\ 1.9,\ 0.4,\ 1.5,\ 1.1)$ and $(a, c, b, \alpha) = (-0.4,\ 0.2,\ 0.3,\ 0.5)$, for example for the derivative with respect to c:

$$e_2 = 0.8 + (-0.4)(1.2) - 0.5 - (0.2 + 0.3 \cdot 1.2)(0) = -0.1800,$$
$$e_3 = 1.9 + (-0.4)(0.8) - 0.5 - (0.2 + 0.3 \cdot 0.8)(-0.18) = 1.1592,$$

$$e_4 = 0.4 + (-0.4)(1.9) - 0.5 - (0.2 + 0.3 \cdot 1.9)(1.1592) = -1.7526,$$

and

$$\begin{aligned}
\tfrac{\partial e_2}{\partial c} &= -e_1 - (0.2 + 0.3 \cdot 1.2)(0) = 0, \\
\tfrac{\partial e_3}{\partial c} &= -e_2 - (0.2 + 0.3 \cdot 0.8)\tfrac{\partial e_2}{\partial c} = 0.18 - (0.44)(0) = 0.1800, \\
\tfrac{\partial e_4}{\partial c} &= -e_3 - (0.2 + 0.3 \cdot 1.9)\tfrac{\partial e_3}{\partial c} = -1.1592 - (0.77)(0.18) = -1.2978.
\end{aligned}$$

Continuing up to $t = 6$ gives $\partial e_6/\partial(a, c, b, \alpha) = (1.5916, -2.8099, -3.0381, -0.4683)$; finite-difference values with a shift of $\pm 10^{-6}$ agree to six decimal places.

Relation to familiar models. If $a = 0$ and $b = 0$, the model becomes MA(1) and the recursion is $e_t = X_t - \alpha - c\, e_{t-1}$, which is the recursion used in *conditional sum of squares* estimation for ARIMA models. The recursive multiplier $(c+bX_{t-1})$ also determines invertibility: by the argument of Section 10, the condition is $\mathbb{E} \log |c + bX_t| < 0$, which for $b = 0$ becomes the classical MA(1) invertibility condition $|c| < 1$.

5 Gradient and Hessian matrix

This section starts from the basic concepts, because these two objects are the main tools for finding the parameter values that minimise $Q(\boldsymbol{\theta})$.

5.1 Basic concepts with one parameter

Suppose we want to minimise a function $Q(\theta)$ of one parameter.

- The **first derivative** $Q'(\theta)$ is the slope of the curve Q at the point θ. If $Q'(\theta) > 0$, Q increases when θ is increased; if $Q'(\theta) < 0$, Q decreases when θ is increased. At the minimum the curve is flat, so
$$Q'(\hat{\theta}) = 0 \qquad \text{(first-order condition)}.$$
- The **second derivative** $Q''(\theta)$ is the curvature of the curve, i.e. how fast the slope changes. A point with $Q' = 0$ is a minimum if the curve bends upwards:
$$Q''(\hat{\theta}) > 0 \qquad \text{(second-order condition)}.$$

A familiar example: the sample mean. For data $y_1, \ldots, y_n$, take $Q(\theta) = \sum_t (y_t - \theta)^2$. Then

$$Q'(\theta) = \sum_t 2(y_t - \theta)(-1) = -2 \sum_t (y_t - \theta), \qquad Q''(\theta) = 2n.$$

The condition $Q'(\hat{\theta}) = 0$ gives $\sum_t y_t - n\hat{\theta} = 0$, i.e. $\hat{\theta} = \bar{y}$; since $Q'' = 2n > 0$, this point is indeed a minimum. So the sample mean is the least squares estimator of θ.

5.2 Many parameters: the gradient vector and the Hessian matrix

When there are q parameters $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_q)'$, the roles of the first and second derivatives are taken by:

- The **gradient vector**, the $q \times 1$ vector of first partial derivatives:

$$\mathbf{g}(\boldsymbol{\theta}) = \frac{\partial Q}{\partial \boldsymbol{\theta}} = \begin{pmatrix} \partial Q/\partial \theta_1 \\ \vdots \\ \partial Q/\partial \theta_q \end{pmatrix}.$$

  The gradient points in the direction of steepest increase of Q; the direction $-\mathbf{g}$ is the direction of steepest descent.

- The **Hessian matrix**, the $q \times q$ matrix of all second partial derivatives:

$$\mathbf{H}(\boldsymbol{\theta}) = \frac{\partial^2 Q}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}'} = \begin{pmatrix} \frac{\partial^2 Q}{\partial \theta_1^2} & \frac{\partial^2 Q}{\partial \theta_1 \partial \theta_2} & \cdots & \frac{\partial^2 Q}{\partial \theta_1 \partial \theta_q} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 Q}{\partial \theta_q \partial \theta_1} & \frac{\partial^2 Q}{\partial \theta_q \partial \theta_2} & \cdots & \frac{\partial^2 Q}{\partial \theta_q^2} \end{pmatrix}.$$

  The (u, v)-th entry describes how the slope with respect to θ_u changes when θ_v changes. The matrix is symmetric because the order of differentiation does not matter ($\partial^2 Q/\partial\theta_u \partial\theta_v = \partial^2 Q/\partial\theta_v \partial\theta_u$).

The conditions for a minimum become:

$$\mathbf{g}(\hat{\boldsymbol{\theta}}) = \mathbf{0} \quad \text{(all slopes zero)}, \qquad \mathbf{H}(\hat{\boldsymbol{\theta}}) \text{ positive definite},$$

where *positive definite* means $\mathbf{v}'\mathbf{H}\mathbf{v} > 0$ for every vector $\mathbf{v} \neq \mathbf{0}$, i.e. the curve Q bends upwards in every direction.

5.3 Example: linear regression and the normal equations

In the linear regression $y_t = \mathbf{x}_t'\boldsymbol{\beta} + e_t$, the sum of squared errors is

$$Q(\boldsymbol{\beta}) = \sum_t (y_t - \mathbf{x}_t'\boldsymbol{\beta})^2 = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}).$$

The error $e_t = y_t - \mathbf{x}_t'\boldsymbol{\beta}$ has derivative $\partial e_t/\partial \boldsymbol{\beta} = -\mathbf{x}_t$. The gradient and Hessian are

$$\mathbf{g}(\boldsymbol{\beta}) = \sum_t 2e_t(-\mathbf{x}_t) = -2\mathbf{X}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}), \qquad \mathbf{H} = 2\sum_t \mathbf{x}_t\mathbf{x}_t' = 2\mathbf{X}'\mathbf{X}.$$

The condition $\mathbf{g} = \mathbf{0}$ gives the normal equations $\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}$, so $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$, the OLS estimator. The Hessian $2\mathbf{X}'\mathbf{X}$ is positive definite when $\mathbf{X}$ has full column rank, so this point is a minimum.

In linear regression the gradient is linear in $\boldsymbol{\beta}$, so $\mathbf{g} = \mathbf{0}$ can be solved directly. In the bilinear model, $e_t(\boldsymbol{\theta})$ is *not* linear in $\boldsymbol{\theta}$ (because e_{t-j} itself depends on $\boldsymbol{\theta}$), so $\mathbf{g}(\boldsymbol{\theta}) = \mathbf{0}$ cannot be solved directly and must be approached iteratively (Section 6).

Connection with the likelihood function. In statistical terms, the gradient of the log-likelihood is called the *score function*, and the negative Hessian of the log-likelihood is called the *observed Fisher information*. Since $\log L = \text{constant} - Q(\boldsymbol{\theta})/(2\sigma^2)$ by (6), the score function is $-\mathbf{g}/(2\sigma^2)$ and the observed information is $\mathbf{H}/(2\sigma^2)$. So minimising Q with the gradient and Hessian is the same as maximising the log-likelihood with the score function and the Fisher information.

5.4 The gradient for the bilinear model

The objective function is $Q(\boldsymbol{\theta}) = \sum_{t=\gamma+1}^{N} e_t(\boldsymbol{\theta})^2$. The partial derivative with respect to θ_u is the sum of the derivatives of each term. By the chain rule, $\dfrac{\partial}{\partial \theta_u}\big[e_t(\boldsymbol{\theta})\big]^2 = 2\, e_t(\boldsymbol{\theta})\, \dfrac{\partial e_t(\boldsymbol{\theta})}{\partial \theta_u} = 2 e_t \psi_{t,u}$. Hence

$$g_u = \frac{\partial Q}{\partial \theta_u} = 2 \sum_{t=\gamma+1}^{N} e_t\, \psi_{t,u}, \qquad u = 1, \ldots, q.$$

To write this in matrix form, stack the errors and their derivatives:

$$\mathbf{e} = \begin{pmatrix} e_{\gamma+1} \\ e_{\gamma+2} \\ \vdots \\ e_N \end{pmatrix}_{n\times 1}, \qquad \mathbf{J} = \begin{pmatrix} \psi_{\gamma+1,1} & \psi_{\gamma+1,2} & \cdots & \psi_{\gamma+1,q} \\ \psi_{\gamma+2,1} & \psi_{\gamma+2,2} & \cdots & \psi_{\gamma+2,q} \\ \vdots & \vdots & & \vdots \\ \psi_{N,1} & \psi_{N,2} & \cdots & \psi_{N,q} \end{pmatrix}_{n\times q}.$$

Row t of $\mathbf{J}$ contains the derivatives of e_t with respect to all parameters; column u contains the derivatives of all errors with respect to θ_u. The u-th entry of $\mathbf{J}'\mathbf{e}$ is column u of $\mathbf{J}$ times $\mathbf{e}$, i.e. $\sum_t \psi_{t,u} e_t$. Therefore

$$\boxed{\mathbf{g} = 2\mathbf{J}'\mathbf{e}.} \tag{13}$$

Compare with linear regression: there $\psi_t = -\mathbf{x}_t$, so $\mathbf{J} = -\mathbf{X}$ and $2\mathbf{J}'\mathbf{e} = -2\mathbf{X}'\mathbf{e}$, the same as the gradient above.

5.5 The Hessian for the bilinear model

The (u, v)-th entry is obtained by differentiating $g_u = 2\sum_t e_t \psi_{t,u}$ with respect to θ_v. Each term $e_t \psi_{t,u}$ is a product of two functions of $\boldsymbol{\theta}$, so by the product rule:

$$\frac{\partial^2 Q}{\partial \theta_u \partial \theta_v} = 2 \sum_t \Big(\frac{\partial e_t}{\partial \theta_v}\, \psi_{t,u} + e_t\, \frac{\partial \psi_{t,u}}{\partial \theta_v} \Big) = \underbrace{2 \sum_t \psi_{t,u} \psi_{t,v}}_{\text{part I}} + \underbrace{2 \sum_t e_t \frac{\partial^2 e_t}{\partial \theta_u \partial \theta_v}}_{\text{part II}}.$$

In matrix form, part I is $2\mathbf{J}'\mathbf{J}$ (the (u, v)-th entry of $\mathbf{J}'\mathbf{J}$ is column u times column v, i.e. $\sum_t \psi_{t,u}\psi_{t,v}$).

Check with linear regression. There e_t is linear in $\boldsymbol{\beta}$, so its second derivatives are zero and part II vanishes. The Hessian $= 2\mathbf{J}'\mathbf{J} = 2\mathbf{X}'\mathbf{X}$, the same as before.

Why part II can be ignored in the bilinear model. In the bilinear model the second derivatives of e_t are not zero, and computing them requires an additional, complicated recursion. However, at the true parameter values e_t is the true error, which is independent of the past, while $\partial^2 e_t / \partial\theta_u \partial\theta_v$ depends only on past data (the right-hand side of (10) contains only quantities at times $t-1$ and earlier). For two independent random variables, the expectation of the product equals the product of the expectations, and $\mathbb{E}(e_t) = 0$, so

$$\mathbb{E}\Big(e_t\, \frac{\partial^2 e_t}{\partial \theta_u \partial \theta_v} \Big) = \mathbb{E}(e_t)\, \mathbb{E}\Big(\frac{\partial^2 e_t}{\partial \theta_u \partial \theta_v} \Big) = 0.$$

By the law of large numbers, $\frac{1}{n}\times$part II tends to zero for large n, while $\frac{1}{n}\times$part I tends to a positive value. The Hessian is therefore approximated by part I alone:

$$\boxed{\mathbf{H} \approx 2\mathbf{J}'\mathbf{J}.} \tag{14}$$

This approximation is widely used in nonlinear least squares (Marquardt, 1963).

An important property. For any vector $\mathbf{v}$, $\mathbf{v}'(\mathbf{J}'\mathbf{J})\mathbf{v} = (\mathbf{J}\mathbf{v})'(\mathbf{J}\mathbf{v}) = \|\mathbf{J}\mathbf{v}\|^2 \geq 0$. So $2\mathbf{J}'\mathbf{J}$ is always positive semi-definite, and positive definite when the columns of $\mathbf{J}$ are linearly independent. Consequently the step direction it produces (Section 6) is always a descent direction for Q, unlike the full Hessian, which may fail to be positive definite far from the minimum.

5.6 Numerical example

In the hand-computed example (Section 13), the model $X_t + aX_{t-1} = \alpha + bX_{t-1}e_{t-1} + e_t$ evaluated at $(a, b, \alpha) = (-0.4, 0.3, 0.5)$ gives

$$\mathbf{g} = \begin{pmatrix} -4.9762 \\ 10.5021 \\ -1.0149 \end{pmatrix}, \qquad \mathbf{H} \approx 2\mathbf{J}'\mathbf{J} = \begin{pmatrix} 12.6208 & -12.0234 & -7.0208 \\ -12.0234 & 18.7351 & 2.9084 \\ -7.0208 & 2.9084 & 6.2091 \end{pmatrix}.$$

The gradient is not zero, so this point is not yet the minimum. The component for b is positive (10.50), meaning that Q will decrease if b is made smaller; the component for a is negative, meaning that Q will decrease if a is made larger. The Hessian determines how far to step in that direction.

6 Newton–Raphson and Gauss–Newton iteration

The previous section gave the gradient $\mathbf{g}$ and the Hessian matrix $\mathbf{H}$. This section explains how they are used to *find* $\hat{\boldsymbol{\theta}}$ numerically, starting from the basic concepts.

6.1 Why iteration is needed

In linear regression, the condition $\mathbf{g}(\boldsymbol{\beta}) = \mathbf{0}$ is a linear system of equations (the normal equations), so $\hat{\boldsymbol{\beta}}$ can be computed directly with a single formula. In the bilinear model, $\mathbf{g}(\boldsymbol{\theta}) = 2\mathbf{J}'\mathbf{e}$ depends on $\boldsymbol{\theta}$ nonlinearly, because e_t and $\psi_{t,u}$ are computed by recursions that involve $\boldsymbol{\theta}$. The equation $\mathbf{g}(\boldsymbol{\theta}) = \mathbf{0}$ cannot be solved with a single formula. What can be done is: start from a guess, improve the guess a little at a time, and repeat until the guess no longer changes. Such a repeated procedure is called an *iteration*.

Iteration notation

- $s = 0, 1, 2, \ldots$ is the **iteration number** (the order of the improvements), *not* time. The time index is still written t.
- $\boldsymbol{\theta}^{(0)}$ is the starting value (the first guess), and $\boldsymbol{\theta}^{(s)}$ is the parameter estimate after s improvements. The parentheses in the superscript indicate the iteration number, not a power.
- $\mathbf{e}^{(s)}$, $\mathbf{J}^{(s)}$, $\mathbf{g}^{(s)}$, $\mathbf{H}^{(s)}$ are the errors, derivative matrix, gradient, and Hessian computed at $\boldsymbol{\theta} = \boldsymbol{\theta}^{(s)}$.
- $\boldsymbol{\delta}^{(s)}$ is the $q \times 1$ **correction vector**: the change in each parameter at iteration s, so that $\boldsymbol{\theta}^{(s+1)} = \boldsymbol{\theta}^{(s)} - \boldsymbol{\delta}^{(s)}$. Its formula is derived below.
- $Q^{(s)} = Q(\boldsymbol{\theta}^{(s)})$ is the sum of squared errors at iteration s; a good iteration gives $Q^{(0)} > Q^{(1)} > Q^{(2)} > \cdots$.

6.2 Taylor series: approximating a function by a polynomial

A Taylor series states that a smooth function $f(\theta)$ near a point θ_0 can be approximated by a polynomial built from its derivatives at that point:

$$f(\theta) \approx f(\theta_0) + f'(\theta_0)(\theta - \theta_0) + \tfrac{1}{2} f''(\theta_0)(\theta - \theta_0)^2 + \tfrac{1}{6} f'''(\theta_0)(\theta - \theta_0)^3 + \cdots$$

Using only the first two terms gives the *first-order approximation* (the tangent line). Using three terms gives the *second-order approximation* (a parabola). The approximation becomes more accurate as θ gets closer to θ_0.

Example. For $f(\theta) = e^\theta$ near $\theta_0 = 1$ (all its derivatives equal $e^1 = 2.7183$), at $\theta = 1.2$:

$$\begin{aligned} \text{true value:} \quad & e^{1.2} = 3.3201, \\ \text{first order:} \quad & e + e(0.2) = e(1 + 0.2) = 3.2619, \\ \text{second order:} \quad & e + e(0.2) + \tfrac{1}{2} e(0.2)^2 = e(1 + 0.2 + 0.02) = 3.3163. \end{aligned}$$

The second-order approximation is much closer to the true value.

Why second order. We want to find a minimum. The first-order approximation is a straight line, which has no minimum (it always increases or always decreases). The second-order approximation is a parabola, which has exactly one minimum when its curvature is positive, and the minimum of a parabola is easy to compute. Third or higher orders are not used because they require higher derivatives and their minimum cannot be computed with a simple formula. So second order is the simplest approximation that still has a minimum.

6.3 The Newton–Raphson method with one parameter

The idea: near the current guess $\theta^{(s)}$, replace $Q(\theta)$ by its second-order Taylor parabola, and take the minimum of that parabola as the next guess. Write $d = \theta - \theta^{(s)}$. The parabola is

$$P(d) = Q(\theta^{(s)}) + Q'(\theta^{(s)})\, d + \tfrac{1}{2} Q''(\theta^{(s)})\, d^2.$$

Differentiate with respect to d and set to zero: $Q'(\theta^{(s)}) + Q''(\theta^{(s)})\, d = 0$, so $d = -Q'(\theta^{(s)})/Q''(\theta^{(s)})$. Then

$$\theta^{(s+1)} = \theta^{(s)} - \frac{Q'(\theta^{(s)})}{Q''(\theta^{(s)})}. \tag{15}$$

Because the parabola is only an approximation, the new point is not necessarily the true minimum; but it is usually closer. The step is repeated until $Q'(\theta^{(s)}) \approx 0$.

Numerical example. Minimise $Q(\theta) = e^\theta - 2\theta$. Its derivatives are $Q'(\theta) = e^\theta - 2$ and $Q''(\theta) = e^\theta$; the true minimum is at $\theta = \ln 2 = 0.693147$. Starting from $\theta^{(0)} = 0$:

s	$\theta^{(s)}$	$Q'(\theta^{(s)})$	$Q''(\theta^{(s)})$	correction Q'/Q''
0	0.000000	−1.000000	1.000000	−1.000000
1	1.000000	0.718282	2.718282	0.264241
2	0.735759	0.087065	2.087065	0.041717
3	0.694042	0.001791	2.001791	0.000895
4	0.693148	0.000001	2.000001	0.000000

First row: $\theta^{(1)} = 0 - (-1)/1 = 1$. Second row: $\theta^{(2)} = 1 - 0.718282/2.718282 = 0.735759$. After four iterations, $\theta^{(s)}$ equals $\ln 2$ to six decimal places.

6.4 Newton–Raphson with many parameters

For a vector $\boldsymbol{\theta}$, the second-order Taylor series around $\boldsymbol{\theta}^{(s)}$ with $\mathbf{d} = \boldsymbol{\theta} - \boldsymbol{\theta}^{(s)}$ is

$$Q(\boldsymbol{\theta}^{(s)} + \mathbf{d}) \approx Q(\boldsymbol{\theta}^{(s)}) + \mathbf{g}'\mathbf{d} + \tfrac{1}{2}\,\mathbf{d}'\mathbf{H}\,\mathbf{d},$$

with $\mathbf{g} = \mathbf{g}^{(s)}$ and $\mathbf{H} = \mathbf{H}^{(s)}$. This is a direct extension of the one-parameter case: $\mathbf{g}'\mathbf{d} = \sum_u g_u d_u$ replaces $Q'd$, and $\mathbf{d}'\mathbf{H}\mathbf{d} = \sum_u \sum_v H_{uv} d_u d_v$ replaces $Q''d^2$.

Differentiate with respect to $\mathbf{d}$. The derivative of $\mathbf{g}'\mathbf{d}$ with respect to $\mathbf{d}$ is $\mathbf{g}$, and the derivative of $\frac{1}{2}\mathbf{d}'\mathbf{H}\mathbf{d}$ is $\mathbf{H}\mathbf{d}$ (because $\mathbf{H}$ is symmetric). Set to zero:

$$\mathbf{g} + \mathbf{H}\mathbf{d} = \mathbf{0} \quad \Longrightarrow \quad \mathbf{d} = -\mathbf{H}^{-1}\mathbf{g}.$$

Define the correction vector $\boldsymbol{\delta}^{(s)} = \mathbf{H}^{-1}\mathbf{g}$ (of size $q \times 1$). The Newton–Raphson step is then

$$\boldsymbol{\theta}^{(s+1)} = \boldsymbol{\theta}^{(s)} + \mathbf{d} = \boldsymbol{\theta}^{(s)} - \boldsymbol{\delta}^{(s)}, \qquad \boldsymbol{\delta}^{(s)} = \left(\mathbf{H}^{(s)}\right)^{-1}\mathbf{g}^{(s)}, \tag{16}$$

which for $q = 1$ reduces to (15).

6.5 The Gauss–Newton method

Newton–Raphson requires the full Hessian. In the bilinear model the Hessian is replaced by its approximation $\mathbf{H} \approx 2\mathbf{J}'\mathbf{J}$ from (14). Substituting $\mathbf{g} = 2\mathbf{J}'\mathbf{e}$ and $\mathbf{H} = 2\mathbf{J}'\mathbf{J}$ into (16), the factors of 2 cancel:

$$\boxed{\boldsymbol{\theta}^{(s+1)} = \boldsymbol{\theta}^{(s)} - \boldsymbol{\delta}^{(s)}, \qquad \boldsymbol{\delta}^{(s)} = \left(\mathbf{J}^{(s)\prime}\mathbf{J}^{(s)}\right)^{-1}\mathbf{J}^{(s)\prime}\mathbf{e}^{(s)}.} \tag{17}$$

Newton–Raphson with this approximate Hessian is called the *Gauss–Newton* method. The size of each object: $\mathbf{J}^{(s)}$ is $n \times q$, $\mathbf{J}^{(s)\prime}\mathbf{J}^{(s)}$ is $q \times q$, $\mathbf{J}^{(s)\prime}\mathbf{e}^{(s)}$ is $q \times 1$, so $\boldsymbol{\delta}^{(s)}$ is $q \times 1$.

Interpretation as a regression. The formula $\boldsymbol{\delta}^{(s)} = (\mathbf{J}'\mathbf{J})^{-1}\mathbf{J}'\mathbf{e}$ is exactly the OLS estimator $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ with $\mathbf{y} = \mathbf{e}$ and $\mathbf{X} = \mathbf{J}$. So **each iteration is one OLS regression**: the errors e_t are the response variable, and the derivatives $\psi_{t,1}, \ldots, \psi_{t,q}$ are the explanatory variables, with no constant. The regression coefficients are $\boldsymbol{\delta}^{(s)}$, which are then subtracted from $\boldsymbol{\theta}^{(s)}$. The same can be seen from linearising the errors: $e_t(\boldsymbol{\theta}) \approx e_t(\boldsymbol{\theta}^{(s)}) + \boldsymbol{\psi}_t'\mathbf{d}$, so $Q \approx \|\mathbf{e} + \mathbf{J}\mathbf{d}\|^2$, which is minimised at $\mathbf{d} = -(\mathbf{J}'\mathbf{J})^{-1}\mathbf{J}'\mathbf{e}$.

Check with linear regression. If the true model is linear, $\mathbf{J} = -\mathbf{X}$ is fixed and Gauss–Newton gives $\boldsymbol{\beta}^{(1)} = \boldsymbol{\beta}^{(0)} + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}^{(0)}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$: the OLS estimator is reached in one iteration from any starting value. The bilinear model needs several iterations because $\mathbf{J}$ changes every time $\boldsymbol{\theta}$ changes.

6.6 Step-size control

The Taylor parabola is accurate only near $\boldsymbol{\theta}^{(s)}$. If $\boldsymbol{\delta}^{(s)}$ is large, the new point may fall in a region where the approximation is poor, so that Q actually increases or recursion (3) explodes (the values of e_t become huge). The step size therefore needs to be controlled. Two common approaches:

- **Step-halving (a simple line search).** Compute $\boldsymbol{\theta}^{(s)} - \lambda\boldsymbol{\delta}^{(s)}$ with $\lambda = 1$. If Q does not decrease or is not finite, repeat with $\lambda = \frac{1}{2}$, then $\frac{1}{4}$, and so on, until Q decreases. The direction stays the same; only its length is shortened.
- **Levenberg–Marquardt (damping).** Solve $(\mathbf{J}'\mathbf{J} + \mu\,\mathrm{diag}(\mathbf{J}'\mathbf{J}))\boldsymbol{\delta} = \mathbf{J}'\mathbf{e}$ with $\mu \geq 0$. For $\mu = 0$ the step is the Gauss–Newton step; for large μ the step is short and points along $-\mathbf{g}$ (the steepest-descent direction). μ is increased if Q rises and decreased if Q falls.

Since $\mathbf{J}'\mathbf{J}$ is positive semi-definite (Section 5), the direction $-\boldsymbol{\delta}^{(s)}$ is always a descent direction, so with a short enough step Q is guaranteed to decrease.

6.7 Stopping rule

The iteration stops when any of the following holds:

1. the relative change in Q is very small: $|Q^{(s)} - Q^{(s+1)}|/Q^{(s)} < \varepsilon$, e.g. $\varepsilon = 10^{-8}$;
2. the correction vector is very small: $\max_u |\delta_u^{(s)}| < \varepsilon_\theta$;
3. the number of iterations reaches a limit $s_{\max}$ (e.g. 100); in this case the result must be checked because it has not necessarily converged.

The authors' original implementation. Subba Rao & Gabr (1984) include the computer program they used. That program uses the gradient without the factor 2 (i.e. $\mathbf{J}'\mathbf{e}$) together with $\mathbf{H} = 2\mathbf{J}'\mathbf{J}$, so its correction is $\frac{1}{2}(\mathbf{J}'\mathbf{J})^{-1}\mathbf{J}'\mathbf{e}$: always a *half* Gauss–Newton step. The program also divides the data by 100 before estimation, stops when the change in $\hat{\sigma}^2$ is less than 10^{-5} or after 20 iterations, and declares the estimation failed if the sum of squared errors exceeds 10^{50}. The half step reaches the same point as the full step, only more slowly (Figure 1).

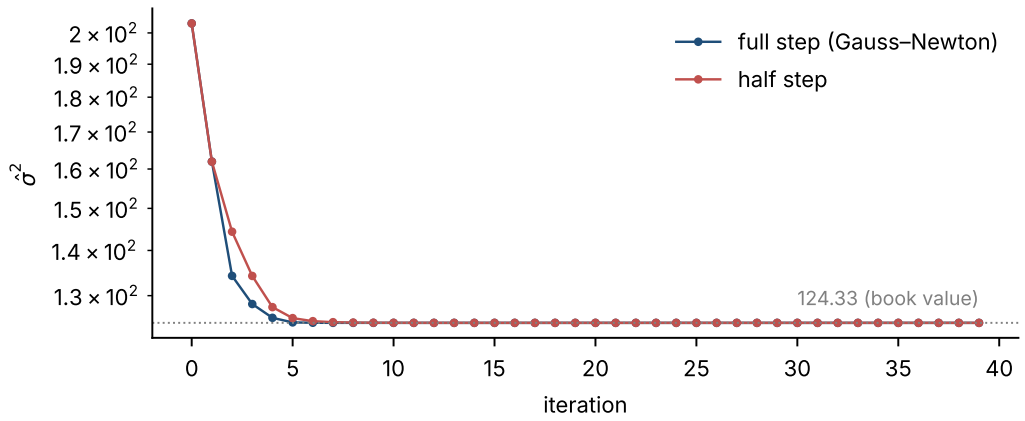


Figure 1: Convergence of $\hat{\sigma}^2$ for Model S (sunspot data, Section 16), starting from the AR model with all $b = 0$. The full step and the half step end at the same point, $\hat{\sigma}^2 = 124.33$.

6.8 Algorithm

A manual computation of this algorithm for two iterations is given in Section 14.

Algorithm 1: Estimation of a subset bilinear model with fixed structure $(\mathcal{K}, \mathcal{B})$

Input: data $X_1, \dots, X_N$; largest lag γ; AR lags $\mathcal{K}$; bilinear pairs $\mathcal{B}$; starting value $\boldsymbol{\theta}^{(0)}$; tolerance ε

Output: $\hat{\boldsymbol{\theta}}$, $\hat{\sigma}^2$, matrix $\mathbf{J}$, residuals $\hat{e}_t$

$s \leftarrow 0$;
repeat
  Set $e_t = 0$ and $\psi_{t,u} = 0$ for $t \le \gamma$;
  for $t = \gamma + 1, \dots, N$ **do**
    compute e_t by (3) and $\psi_{t,u}$ by (10), using $\boldsymbol{\theta}^{(s)}$;
  $Q^{(s)} \leftarrow \sum_t e_t^2$; $\quad \boldsymbol{\delta}^{(s)} \leftarrow (\mathbf{J}'\mathbf{J})^{-1}\mathbf{J}'\mathbf{e}$;
  $\lambda \leftarrow 1$;
  while $Q(\boldsymbol{\theta}^{(s)} - \lambda\boldsymbol{\delta}^{(s)})$ *is not finite* ***or*** $> Q^{(s)}$ **do**
    $\lambda \leftarrow \lambda/2$; `// step-halving`
  $\boldsymbol{\theta}^{(s+1)} \leftarrow \boldsymbol{\theta}^{(s)} - \lambda\boldsymbol{\delta}^{(s)}$; $\quad s \leftarrow s + 1$;
until $|Q^{(s-1)} - Q^{(s)}|/Q^{(s-1)} < \varepsilon$ ***or*** $s = s_{\max}$;
$\hat{\boldsymbol{\theta}} \leftarrow \boldsymbol{\theta}^{(s)}$; $\quad \hat{\sigma}^2 \leftarrow Q(\hat{\boldsymbol{\theta}})/n$;

7 Starting values

The function $Q(\boldsymbol{\theta})$ may have more than one local minimum. The iteration stops at the minimum "nearest" to the starting point, so the starting values largely determine the result.

7.1 From the autoregressive model

A natural choice is to start from the model without bilinear terms. Fit the AR model with lags $\mathcal{K}$ and a constant,

$$X_t = -\sum_{k \in \mathcal{K}} a_k X_{t-k} + \alpha + e_t,$$

by ordinary OLS (regress X_t on $-X_{t-k}$ and a column of ones, for $t = \gamma + 1, \dots, N$), then set all $b_{ij} = 0$.

If the AR model is fitted to the centred data $x_t = X_t - \bar{X}$ (without a constant), the constant for the original data is obtained by substituting $x_t = X_t - \bar{X}$ into $x_t + \sum a_k x_{t-k} = e_t$:

$$X_t - \bar{X} + \sum a_k (X_{t-k} - \bar{X}) = e_t \implies X_t + \sum a_k X_{t-k} = \bar{X}\Big(1 + \sum_{k \in \mathcal{K}} a_k\Big) + e_t,$$

so that $\alpha = \bar{X}(1 + \sum a_k)$.

7.2 The repeated-residual method

Idea: if e_{t-j} is treated as *known*, model (2) becomes linear in all parameters, because $X_{t-i}e_{t-j}$ is just an explanatory variable. Then:

1. start from the residuals of some initial model (e.g. the AR residuals);
2. regress X_t on $\{-X_{t-k}\}$, $\{X_{t-i}e_{t-j}\}$, and 1 by OLS, keeping the residuals fixed;

3. recompute e_t by (3) using the new parameters;
4. repeat a few times (e.g. at most 5) or until $\hat{\sigma}^2$ hardly changes.

The result can be used as the starting value for Algorithm 1. This method is not guaranteed to decrease Q: for Model S, five rounds actually increase $\hat{\sigma}^2$ to about 7360, although Gauss–Newton afterwards still reaches 124.33.

7.3 Staged starting values

When the model is built by adding bilinear terms one at a time (Section 8), the starting values for the larger model are taken from the previous model: all old parameters use the estimates from the previous model, and the coefficient of the new term starts at 0. This strategy turns out to be important. For Model L1 (Section 16), estimating all six bilinear terms at once from the AR starting values ends at a local minimum with $\hat{\sigma}^2 = 0.0256$, whereas adding them in stages reaches $\hat{\sigma}^2 = 0.0223$.

7.4 Multiple starting points (multi-start)

To guard against local minima, run Algorithm 1 from several starting values for the coefficients b (e.g. $b^{(0)} \in \{-0.4, -0.2, 0, 0.2, 0.4\}$, adjusted to the scale of the data) and take the result with the smallest Q. Section 15 shows that this simple step removes almost all failures.

8 Model selection

8.1 Akaike's information criterion

AIC is defined as $-2\log L(\hat{\boldsymbol{\theta}}, \hat{\sigma}^2) + 2q$, where q is the number of parameters. From (8),

$$-2\log L(\hat{\boldsymbol{\theta}}, \hat{\sigma}^2) = n(1 + \log 2\pi) + n\log\frac{Q(\hat{\boldsymbol{\theta}})}{n} = n\log\hat{\sigma}^2 + n(1 + \log 2\pi).$$

The term $n(1 + \log 2\pi)$ is the same for all models on the same data, so it can be dropped. This gives

$$\boxed{\mathrm{AIC} = n\log\hat{\sigma}^2 + 2q, \qquad \mathrm{NAIC} = \frac{\mathrm{AIC}}{n} = \log\hat{\sigma}^2 + \frac{2q}{n}.} \tag{18}$$

NAIC (normalised AIC) is the form reported by Subba Rao & Gabr (1984). The number of parameters q includes the constant α (or the mean, for an AR model on centred data), but does not include σ^2. Since σ^2 is in every model, counting it or not does not change the ranking of the models. The model with the smallest AIC is chosen.

8.2 Stepwise selection procedure

1. **Fix** γ (the largest lag) and $n = N - \gamma$.
2. **Full AR order.** Fit AR models of order $1, 2, \ldots, \gamma$ by OLS on observations $t = \gamma+1, \ldots, N$; choose the order p with the smallest AIC.

Table 1: Checking formula (18) against the figures reported by Subba Rao & Gabr (1984). The models are described in Section 16.

Data	Model	$\hat{\sigma}^2$	q	n	NAIC computed	NAIC reported
Sunspot	AR(9)	199.27	10	211	5.389	5.39
Sunspot	subset AR lags $\{1, 2, 9\}$	203.21	4	211	5.352	5.352
Sunspot	Model S	124.33	11	211	4.927	4.927
Log lynx	Model L1	0.0223	13	88	-3.508	-3.508

3. **Subset AR.** Among all non-empty subsets of $\{1, \ldots, p\}$ (there are $2^p - 1$), choose the one with the smallest AIC. For $p \leq 12$ this is at most 4095 OLS regressions, so an exhaustive search is feasible. For the sunspot and log lynx data, the exhaustive search gives the same subsets as those reported by Subba Rao & Gabr (1984), who used a different search algorithm.

4. **First bilinear term.** Fix ξ with $p \leq \xi \leq \gamma$ and the candidate set $T_1 = \{(i, j) : 1 \leq i, j \leq \xi\}$. For each $(i, j) \in T_1$, fit the subset AR model plus one term $b_{ij} X_{t-i} e_{t-j}$ with Algorithm 1; choose the pair (r_1, s_1) with the smallest AIC.

5. **Next terms.** Repeat with candidates $T_2 = T_1 \setminus \{(r_1, s_1)\}$, using staged starting values, and so on.

6. **Stop** when adding a term no longer decreases AIC.

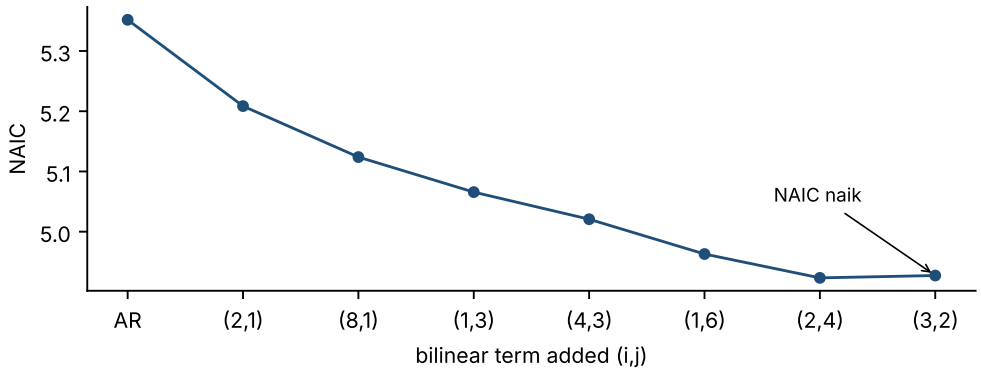


Figure 2: The NAIC path as the bilinear terms of Model S are added one at a time in the order reported by Subba Rao & Gabr (1984). The last term, $(3, 2)$, raises NAIC from 4.9233 to 4.9272; the t-value of its coefficient is only 1.2.

Note

Re-running the stepwise procedure with $\xi = 9$ on the sunspot data selects the first six terms of Model S in exactly the same order: $(2, 1), (8, 1), (1, 3), (4, 3), (1, 6), (2, 4)$. At the seventh step, the procedure selects $(8, 5)$ with $\hat{\sigma}^2 = 120.2$, whereas Model S contains $(3, 2)$. Figure 2 shows that the term $(3, 2)$ actually raises NAIC. So the stopping rule used in the original report is not entirely consistent. A strict rule is recommended: stop when AIC increases, and reject candidates that are not invertible (Section 10).

9 Standard errors

9.1 Derivation

Near the estimate $\hat{\boldsymbol{\theta}}$, linearising the errors as in Section 6 gives the approximate regression model

$$\mathbf{e}(\boldsymbol{\theta}_0) \approx \mathbf{e}(\hat{\boldsymbol{\theta}}) + \mathbf{J}(\boldsymbol{\theta}_0 - \hat{\boldsymbol{\theta}}),$$

where $\boldsymbol{\theta}_0$ is the true parameter and $\mathbf{e}(\boldsymbol{\theta}_0)$ are the true i.i.d. $N(0, \sigma^2)$ errors. Rearranging gives $\mathbf{e}(\hat{\boldsymbol{\theta}}) \approx \mathbf{J}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + \mathbf{e}(\boldsymbol{\theta}_0)$, a linear regression equation with "coefficients" $\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0$. Since $\hat{\boldsymbol{\theta}}$ minimises the sum of squares, $\mathbf{J}'\mathbf{e}(\hat{\boldsymbol{\theta}}) = \mathbf{0}$ (zero gradient), and from this

$$\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0 \approx -(\mathbf{J}'\mathbf{J})^{-1}\mathbf{J}'\mathbf{e}(\boldsymbol{\theta}_0).$$

Treating $\mathbf{J}$ as fixed, $\operatorname{Var}(\mathbf{J}'\mathbf{e}(\boldsymbol{\theta}_0)) = \sigma^2\mathbf{J}'\mathbf{J}$, so

$$\boxed{\widehat{\operatorname{Cov}}(\hat{\boldsymbol{\theta}}) = \hat{\sigma}^2(\mathbf{J}'\mathbf{J})^{-1}, \qquad \operatorname{SE}(\hat{\theta}_u) = \sqrt{\left[\hat{\sigma}^2(\mathbf{J}'\mathbf{J})^{-1}\right]_{uu}}.} \tag{19}$$

The same result is obtained from the Fisher information matrix: the second derivative of $-\log L$ with respect to $\boldsymbol{\theta}$ is $\frac{1}{2\sigma^2}\mathbf{H} = \frac{1}{\sigma^2}\mathbf{J}'\mathbf{J}$, and the inverse of the information matrix gives (19). The matrix $\mathbf{J}$ is the same one used in the last iteration.

Limits on the accuracy of the standard errors

The derivation above is an approximation. For general bilinear models, the asymptotic distribution of the estimator has not been established in general; existing results cover only special cases and require strong moment conditions (for example Grahn, 1995, requires $\mathbb{E}X_t^8 < \infty$). Ling, Peng & Zhu also state that a general asymptotic theory for these models is not yet established. The simulations in Section 15 show that (19) is reasonably good for the AR coefficients and the constant, but somewhat too small for b. For formal conclusions, a parametric bootstrap is safer.

9.2 Parameter significance test

This test checks whether a parameter differs significantly from zero, i.e. whether the corresponding term really plays a role in the model.

Hypotheses. For the u-th parameter (a_k, b_{ij}, c_l, or α):

$$H_0 : \ \theta_u = 0 \qquad \text{vs} \qquad H_1 : \ \theta_u \neq 0.$$

H_0 states that the term containing θ_u has no effect on X_t. For example $H_0 : b_{ij} = 0$ means that the interaction $X_{t-i}e_{t-j}$ is not needed. H_1 states that the term has an effect.

Test statistic. With the standard error from (19),

$$t_u = \frac{\hat{\theta}_u - 0}{\operatorname{SE}(\hat{\theta}_u)} = \frac{\hat{\theta}_u}{\sqrt{\left[\hat{\sigma}^2(\mathbf{J}'\mathbf{J})^{-1}\right]_{uu}}} \overset{H_0}{\approx} N(0,1).$$

Its form is the same as the t-test in linear regression: the estimate divided by its standard error. The distribution is only an *approximation* for large n, because the estimator in the bilinear model is nonlinear. For small samples, the t distribution with $n - q$ degrees of freedom is often used as a more cautious approximation; for large n the two are almost the same.

Decision. At significance level α, reject H_0 if $|t_u| > z_{\alpha/2}$ (e.g. 1.96 for $\alpha = 5\%$), or if

$$p\text{-value} = 2\big(1 - \Phi(|t_u|)\big) < \alpha.$$

Equivalently, H_0 is rejected if the $100(1-\alpha)\%$ confidence interval $\hat{\theta}_u \pm z_{\alpha/2}\,\mathrm{SE}(\hat{\theta}_u)$ does not contain zero.

Example (Model S, Section 16).

- $\hat{b}_{2,1} = -0.014575$, SE $= 0.001018$, so $t = -14.3$. Since $|t| > 1.96$, $H_0 : b_{2,1} = 0$ is rejected; the term $X_{t-2}e_{t-1}$ has a significant effect. 95% confidence interval: $-0.014575 \pm 1.96(0.001018) = (-0.0166, -0.0126)$, which does not contain zero.
- $\hat{b}_{3,2} = 0.001781$, SE $= 0.001487$, so $t = 1.20$ and p-value $= 2(1 - \Phi(1.20)) = 0.23$. Since $|t| < 1.96$, $H_0 : b_{3,2} = 0$ is not rejected; there is no evidence that the term $X_{t-3}e_{t-2}$ is needed. This agrees with Figure 2: adding this term actually raises NAIC.

Caution with tests for the bilinear coefficients

The accuracy of this test depends on the accuracy of the standard errors. The simulation study (Section 15) shows that the standard errors for b tend to be too small (the coverage probability of the 95% interval is only 91.5%). As a result, the values of $|t|$ for b tend to be too large, so $H_0 : b_{ij} = 0$ is rejected more often than it should be. For coefficients b whose $|t|$ is close to the critical value, conclusions should be confirmed with a bootstrap confidence interval. For the AR coefficients and the constant, this test is more reliable.

10 Invertibility

10.1 Why it matters

The errors $e_t(\boldsymbol{\theta})$ are computed with the initial values (4), which are in fact wrong. The computed errors only approximate the true errors if the effect of those wrong initial values dies out over time. Granger & Andersen (1978) call a model *invertible* if $\mathbb{E}(e_t - \hat{e}_t)^2 \to 0$ as $t \to \infty$, where $\hat{e}_t$ is the reconstructed error. If the model is not invertible, $Q(\boldsymbol{\theta})$ no longer measures the fit of the model, and minimisation can produce a spuriously small $\hat{\sigma}^2$.

10.2 Propagation equation for the initial-value error

Let e_t be the true error and $\hat{e}_t$ the error computed with different initial values; both satisfy (3) with the same data and parameters:

$$\hat{e}_t = X_t + \sum a_k X_{t-k} - \alpha - \sum b_{ij} X_{t-i}\,\hat{e}_{t-j},$$
$$e_t = X_t + \sum a_k X_{t-k} - \alpha - \sum b_{ij} X_{t-i}\,e_{t-j}.$$

Subtract the second equation from the first. All terms that do not contain errors cancel. With $\delta_t = \hat{e}_t - e_t$:

$$\boxed{\delta_t = -\sum_{(i,j)\in\mathcal{B}} b_{ij} X_{t-i}\,\delta_{t-j}} \tag{20}$$

(plus $-\sum c_j \delta_{t-j}$ if there are MA terms). The model is invertible if $\delta_t \to 0$.

10.3 The case of one bilinear term

For $X_t = bX_{t-i}e_{t-j} + e_t$, equation (20) becomes $\delta_t = -bX_{t-i}\delta_{t-j}$. Applying it repeatedly every j steps,

$$|\delta_t| = |bX_{t-i}| \cdot |bX_{t-i-j}| \cdot |bX_{t-i-2j}| \cdots |\delta_{\text{init}}|, \qquad \log|\delta_t| = \sum_{\ell} \log|bX_{t-i-\ell j}| + \log|\delta_{\text{init}}|.$$

The sum on the right-hand side has about t/j terms. For a stationary and ergodic series, the average of these terms tends to its expectation (the ergodic theorem, the time-series version of the law of large numbers), so $\frac{1}{t}\log|\delta_t| \to \frac{1}{j}\mathbb{E}\log|bX_t|$. Hence $\delta_t \to 0$ if

$$\mathbb{E}\log|b\,X_t| < 0.$$

This condition, involving the expectation of the logarithm of an absolute value, agrees with the result of Quinn (1982) for the one-parameter bilinear model.

10.4 A practical diagnostic

For general models there is no closed-form formula. Use (20) numerically: set $\delta_t = 1$ for $t \le \gamma$, run (20) with the data and $\hat{\boldsymbol{\theta}}$ up to $t = N$, then inspect $\log_{10}|\delta_N|$. A strongly negative value means the effect of the initial values has died out (invertible); a positive value means the effect grows (not invertible).

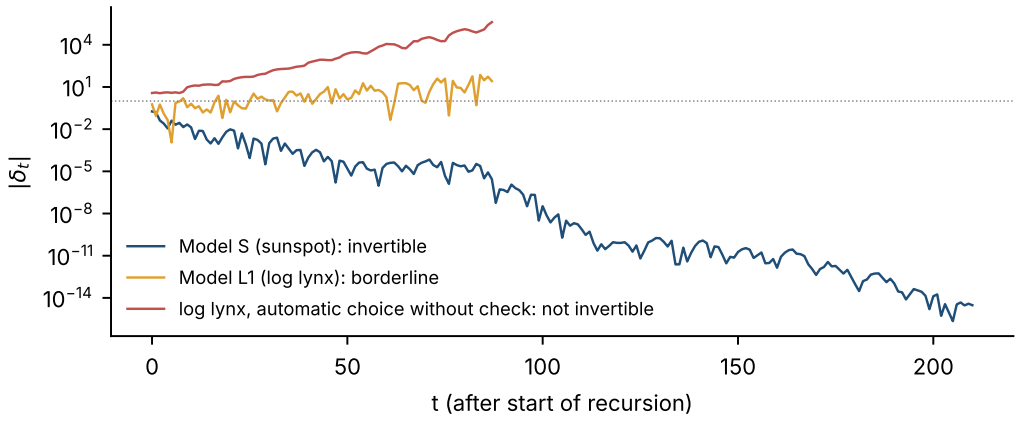


Figure 3: Propagation of the initial-value error (20). Model S is clearly invertible ($|\delta| \approx 10^{-14}$ at the end). Model L1 is borderline ($|\delta|$ grows by a factor of about 70). The model chosen automatically without an invertibility check grows by a factor of about 10^4, even though its $\hat{\sigma}^2$ (0.0114) looks much better than that of Model L1 (0.0223).

11 Residual tests

If the model is adequate, the errors e_t are independent and normally distributed, so the residuals $\hat{e}_t$ should resemble an i.i.d. normal sequence. The two tests below check these two properties. In this section N denotes the number of residuals, and $z_{\alpha/2}$ the upper quantile of the standard normal ($z_{0.025} = 1.96$ for a 5% significance level).

11.1 The independence test W

Residuals may be uncorrelated and yet not independent; such dependence is often visible in their squares (for example, periods with large residuals tend to be followed by large residuals). This test checks the lag-1 autocorrelation of the squared errors. Let $y_t = e_t^2$ with population autocorrelation $\rho_y(1) = \mathrm{Corr}(e_t^2, e_{t+1}^2)$.

Hypotheses.

$$H_0 : \ \rho_y(1) = 0 \qquad \text{vs} \qquad H_1 : \ \rho_y(1) \neq 0.$$

H_0 states that there is no correlation between consecutive squared errors, which certainly holds if the errors are independent. H_1 states that the squared errors are correlated, meaning the errors are not independent: there is structure that the model has not yet captured. Note that failing to reject H_0 does not prove independence; the test only checks one of its consequences.

Test statistic. With $y_t = \hat{e}_t^2$, $\bar{y}$ its mean, and

$$\hat{R}_y(s) = \frac{1}{N}\sum_{t=1}^{N-s}(y_t - \bar{y})(y_{t+s} - \bar{y}), \qquad \hat{\rho}_y(1) = \frac{\hat{R}_y(1)}{\hat{R}_y(0)}, \qquad \hat{R}_y(0,1,1) = \frac{1}{N}\sum_{t=1}^{N-1}(y_t - \bar{y})^2(y_{t+1} - \bar{y})^2.$$

Under H_0, $\hat{R}_y(1)$ is an average of $N-1$ uncorrelated terms $(y_t - \bar{y})(y_{t+1} - \bar{y})$, each with variance $R_y(0,1,1) = \mathbb{E}[(y_t - \mu_y)^2(y_{t+1} - \mu_y)^2]$, so $\mathrm{Var}(\hat{R}_y(1)) \approx R_y(0,1,1)/N$. The test statistic is

$$W = \frac{\hat{\rho}_y(1)\,\hat{R}_y(0)\sqrt{N}}{\sqrt{\hat{R}_y(0,1,1)}} = \frac{\sqrt{N}\,\hat{R}_y(1)}{\sqrt{\hat{R}_y(0,1,1)}} \overset{H_0}{\approx} N(0,1).$$

Decision. Reject H_0 at significance level α if $|W| > z_{\alpha/2}$, or if p-value $= 2\big(1 - \Phi(|W|)\big) < \alpha$, where Φ is the standard normal distribution function.

11.2 Normality test (Lomnicki)

The normal distribution has zero skewness and zero excess kurtosis. For errors with mean μ and standard deviation σ,

$$\gamma_1 = \frac{\mathbb{E}(e_t - \mu)^3}{\sigma^3}, \qquad \gamma_2 = \frac{\mathbb{E}(e_t - \mu)^4}{\sigma^4} - 3.$$

Hypotheses. Two hypotheses are tested separately:

$$H_0 : \ \gamma_1 = 0 \ \text{ vs } \ H_1 : \ \gamma_1 \neq 0 \qquad \text{and} \qquad H_0 : \ \gamma_2 = 0 \ \text{ vs } \ H_1 : \ \gamma_2 \neq 0.$$

In the first test, H_0 states that the error distribution is symmetric like the normal, and H_1 that it is skewed to the right ($\gamma_1 > 0$) or to the left ($\gamma_1 < 0$). In the second test, H_0 states that the tails of the distribution are as heavy as those of the normal, and H_1 that they are heavier ($\gamma_2 > 0$, extreme values occur more often) or lighter ($\gamma_2 < 0$). If either H_0 is rejected, the normality assumption is not supported by the data.

Test statistics. With $m_k = \frac{1}{N}\sum(\hat{e}_t - \bar{e})^k$, the estimators of skewness and excess kurtosis are

$$G_1 = \frac{m_3}{m_2^{3/2}}, \qquad G_2 = \frac{m_4}{m_2^2} - 3.$$

Lomnicki (1961) showed that for a stationary Gaussian series,

$$\mathrm{Var}(G_1) \approx \frac{6}{N}\sum_s \rho_e^2(s), \qquad \mathrm{Var}(G_2) \approx \frac{24}{N}\sum_s \rho_e^4(s),$$

where $\rho_e(s)$ is the residual autocorrelation (for i.i.d. residuals the sums equal 1, so $\mathrm{Var}(G_1) \approx 6/N$ and $\mathrm{Var}(G_2) \approx 24/N$). The test statistics are

$$u_1 = \frac{G_1}{\sqrt{\mathrm{Var}(G_1)}} \overset{H_0}{\approx} N(0,1), \qquad u_2 = \frac{G_2}{\sqrt{\mathrm{Var}(G_2)}} \overset{H_0}{\approx} N(0,1).$$

Decision. Reject $H_0 : \gamma_1 = 0$ if $|u_1| > z_{\alpha/2}$, and reject $H_0 : \gamma_2 = 0$ if $|u_2| > z_{\alpha/2}$. The two tests can be combined: under normality, u_1 and u_2 are asymptotically independent, so $u_1^2 + u_2^2 \approx \chi^2_2$ (the same idea as the Jarque–Bera test); reject normality if $u_1^2 + u_2^2 > \chi^2_{2;\alpha}$ ($= 5.99$ for $\alpha = 5\%$).

Difference from the formula printed in the book

Subba Rao & Gabr (1984) write $\mathrm{Var}(G_1) = 6\sum\rho_e^2(s)$ and $\mathrm{Var}(G_2) = 24\sum\rho_e^4(s)$ without the factor $1/N$, and the values of u_1, u_2 they report can only be reproduced with that formula without $1/N$. The factor $1/N$ should be there, since the sample skewness of N normal observations has variance of about $6/N$.

Table 2: Residual tests on the sunspot data ($N = 211$ residuals).

Model	Statistic	Reported in book	Formula without $1/N$	Formula with $1/N$
AR(9)	W	2.459	2.457	
	u_1	0.333	0.329	4.77
	u_2	0.360	0.361	5.24
Model S	W	1.561	1.561	
	u_1	0.302	0.303	4.40
	u_2	0.200	0.200	2.91

For Model S at the 5% level: $|W| = 1.561 < 1.96$, so $H_0 : \rho_y(1) = 0$ is not rejected. With the correct formula, $|u_1| = 4.40 > 1.96$ and $|u_2| = 2.91 > 1.96$, so both H_0 are rejected: the residuals are skewed to the right ($G_1 = 0.74$) and heavy-tailed. This conclusion differs from that of the book, which used the formula without $1/N$. For the AR(9) model, $|W| = 2.46 > 1.96$, so the independence H_0 is rejected; the bilinear model removes this dependence.

11.3 What to do if a test fails

If the independence H_0 is rejected. Correlated squared residuals mean there is still nonlinear structure (or a changing conditional variance) that the model has not captured. Steps to try:

1. Check the autocorrelation of the squared residuals at several lags jointly (e.g. the Ljung–Box statistic on $\hat{e}_t^2$, known as the McLeod–Li test) to see which lags are problematic. Also check the autocorrelation of the ordinary residuals $\hat{e}_t$; if it is significant, add AR lags.

2. Add bilinear terms: continue the stepwise procedure (Section 8) or increase ξ, especially at the lags indicated by step 1.
3. Check the invertibility of the model (Section 10). In a non-invertible model, the residuals do not reconstruct the errors correctly, so residual tests are meaningless.
4. If adding bilinear terms does not remove the dependence, consider another class of models that models the conditional variance directly, such as ARCH/GARCH, or threshold models.

If the normality H_0 is rejected.

1. Check for outliers: very large residuals may come from data errors or unusual events. Normality is often rejected because of just a few observations.
2. The parameter estimates can still be used as *conditional least squares* estimators, because minimising $Q(\boldsymbol{\theta})$ does not require normality. What is affected is the inference: the standard errors (19), AIC, and the forecast intervals derived from the normal distribution become less accurate.
3. For inference, use the residual bootstrap: generate new data from the fitted model with errors drawn at random (with replacement) from the residuals $\hat{e}_t$, re-estimate the parameters, and use the spread of these estimates for standard errors and confidence intervals.
4. For forecasting by Monte Carlo simulation (Section 12), draw future errors from the residuals $\hat{e}_t$ rather than from $N(0, \hat{\sigma}^2)$.
5. Consider transforming the data (log, square root, or Box–Cox) before modelling. Note that back-transformed forecasts are biased and need to be corrected.

12 Forecasting

The optimal forecast (in the sense of the smallest mean squared error) of X_{t+h} given information up to time t is its conditional expectation, $\hat{X}_t(h) = \mathbb{E}(X_{t+h} \mid \mathcal{F}_t)$. Assuming invertibility, $e_1, \ldots, e_t$ can be computed from the data, whereas $e_{t+1}, e_{t+2}, \ldots$ have not yet occurred, are independent, and have expectation 0 and variance σ^2.

12.1 One step ahead

From (2) at time $t+1$: $X_{t+1} = \mu_{t+1} + e_{t+1}$, where μ_{t+1} contains only quantities up to time t. So

$$\hat{X}_t(1) = \mu_{t+1} = -\sum_k a_k X_{t+1-k} + \alpha + \sum_{(i,j)} b_{ij} X_{t+1-i} e_{t+1-j}, \qquad X_{t+1} - \hat{X}_t(1) = e_{t+1}.$$

So the out-of-sample one-step forecast errors are obtained simply by continuing recursion (3) on the new data with $\hat{\boldsymbol{\theta}}$.

12.2 Two or more steps: derivation on an example

Take $X_t + aX_{t-1} = \alpha + bX_{t-1}e_{t-1} + e_t$. For $h = 2$:

$$X_{t+2} = -aX_{t+1} + \alpha + bX_{t+1}e_{t+1} + e_{t+2}.$$

Take conditional expectations term by term:

- $\mathbb{E}(X_{t+1} \mid \mathcal{F}_t) = \hat{X}_t(1)$ and $\mathbb{E}(e_{t+2} \mid \mathcal{F}_t) = 0$.
- The bilinear term contains X_{t+1} and e_{t+1}, which have both not yet occurred and are *not independent*, because $X_{t+1} = \mu_{t+1} + e_{t+1}$:

$$\mathbb{E}(X_{t+1}e_{t+1} \mid \mathcal{F}_t) = \mathbb{E}\big((\mu_{t+1} + e_{t+1})e_{t+1} \mid \mathcal{F}_t\big) = \mu_{t+1}\mathbb{E}(e_{t+1}) + \mathbb{E}(e_{t+1}^2) = 0 + \sigma^2.$$

So $\hat{X}_t(2) = -a\hat{X}_t(1) + \alpha + b\sigma^2$. In the same way, for $h \geq 2$ in this model $\hat{X}_t(h) = -a\hat{X}_t(h-1) + \alpha + b\sigma^2$. Note that the bilinear term contributes $b\sigma^2$, not zero.

The case where X is later than e. Sometimes $\mathbb{E}(X_{t+2}e_{t+1} \mid \mathcal{F}_t)$ is needed (for example for the term $X_{t+h-1}e_{t+h-2}$ when $h = 3$). Multiply the equation for X_{t+2} by e_{t+1} and take expectations:

$$\mathbb{E}(X_{t+2}e_{t+1} \mid \mathcal{F}_t) = -a\,\mathbb{E}(X_{t+1}e_{t+1}) + \alpha\mathbb{E}(e_{t+1}) + b\,\mathbb{E}(X_{t+1}e_{t+1}^2) + \mathbb{E}(e_{t+2}e_{t+1}).$$

The second and fourth terms are zero. $\mathbb{E}(X_{t+1}e_{t+1}) = \sigma^2$ as above, and $\mathbb{E}(X_{t+1}e_{t+1}^2) = \mathbb{E}((\mu_{t+1} + e_{t+1})e_{t+1}^2) = \mu_{t+1}\sigma^2 + \mathbb{E}(e_{t+1}^3) = \hat{X}_t(1)\sigma^2$ (the third moment of the normal is zero). So $\mathbb{E}(X_{t+2}e_{t+1} \mid \mathcal{F}_t) = -a\sigma^2 + b\sigma^2\hat{X}_t(1)$.

12.3 General rule

For a term $X_{t+h-i}e_{t+h-j}$ in the equation for X_{t+h}, the conditional expectation follows four cases:

Condition	$\mathbb{E}(X_{t+h-i}e_{t+h-j} \mid \mathcal{F}_t)$	Reason
$i \geq h$ and $j \geq h$	$X_{t+h-i}\,e_{t+h-j}$	both have already occurred
$i < h \leq j$	$\hat{X}_t(h-i)\,e_{t+h-j}$	e has occurred, X has not
$j < h$ and $j < i$	0	e has not occurred and is later than X, so independent
$i \leq j < h$	σ^2 if $i = j$; a recursion as in the example above if $i < j$	X depends on e

For models with many terms, the last case produces a long chain of recursions.

12.4 The Monte Carlo alternative

The conditional expectation can be computed without deriving formulas: generate many future paths with $e_{t+1}, \ldots, e_{t+h} \sim N(0, \hat{\sigma}^2)$ and the past data fixed, compute $X_{t+1}, \ldots, X_{t+h}$ with (2), then average X_{t+h}. With many paths, this average approaches the exact $\hat{X}_t(h)$. For Model S, this approach gives the same results as the reported values:

Table 3: Forecast mean squared error $\hat{\sigma}^2(h)$, sunspot data 1921–1955.

h	1	2	3	4	5
Reported by Subba Rao & Gabr (1984)	123.77	337.54	569.79	659.05	718.87
Recomputed (Monte Carlo, 20 000 paths)	123.77*	337.48	570.75	659.00	720.57

* computed exactly from the error recursion; the one-step forecast values are also identical to those reported (24.55, 11.43, 10.08, …).

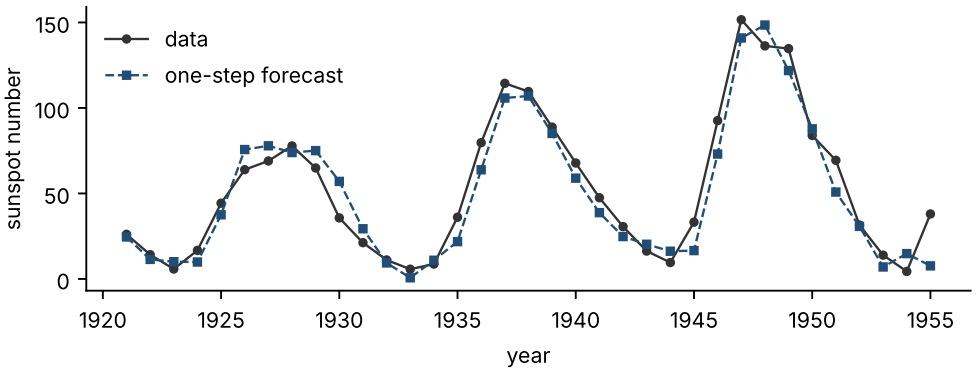


Figure 4: One-step forecasts of Model S for the years 1921–1955.

13 A hand-computed example

Take the model

$$X_t + aX_{t-1} = \alpha + bX_{t-1}e_{t-1} + e_t,$$

with six observations $X = (1.2,\ 0.8,\ 1.9,\ 0.4,\ 1.5,\ 1.1)$, $\gamma = 1$, $n = 5$, and starting value $\boldsymbol{\theta}^{(0)} = (a, b, \alpha) = (-0.4,\ 0.3,\ 0.5)$.

13.1 The error recursion

The formula is $e_t = X_t + aX_{t-1} - \alpha - bX_{t-1}e_{t-1}$ with $e_1 = 0$:

$$\begin{aligned}
e_2 &= 0.8 + (-0.4)(1.2) - 0.5 - (0.3)(1.2)(0) = 0.8 - 0.48 - 0.5 = -0.1800,\\
e_3 &= 1.9 + (-0.4)(0.8) - 0.5 - (0.3)(0.8)(-0.18) = 1.9 - 0.32 - 0.5 + 0.0432 = 1.1232,\\
e_4 &= 0.4 + (-0.4)(1.9) - 0.5 - (0.3)(1.9)(1.1232) = 0.4 - 0.76 - 0.5 - 0.64022 = -1.5002,\\
e_5 &= 1.5 + (-0.4)(0.4) - 0.5 - (0.3)(0.4)(-1.5002) = 1.5 - 0.16 - 0.5 + 0.18003 = 1.0200,\\
e_6 &= 1.1 + (-0.4)(1.5) - 0.5 - (0.3)(1.5)(1.0200) = 1.1 - 0.6 - 0.5 - 0.45901 = -0.4590.
\end{aligned}$$

13.2 Recursive derivatives

As in Section 4, the derivative of the error with respect to each parameter is written with the symbol ψ:

$$\psi_{t,a} = \frac{\partial e_t}{\partial a}, \qquad \psi_{t,b} = \frac{\partial e_t}{\partial b}, \qquad \psi_{t,\alpha} = \frac{\partial e_t}{\partial \alpha}.$$

So $\psi_{t,a}$ measures how much e_t changes when a is changed slightly, and so on. From (10), with all derivatives zero at $t = 1$:

$$\begin{aligned}
\psi_{t,a} &= \frac{\partial e_t}{\partial a} = X_{t-1} - bX_{t-1}\frac{\partial e_{t-1}}{\partial a} = X_{t-1} - bX_{t-1}\psi_{t-1,a},\\
\psi_{t,b} &= \frac{\partial e_t}{\partial b} = -X_{t-1}e_{t-1} - bX_{t-1}\frac{\partial e_{t-1}}{\partial b} = -X_{t-1}e_{t-1} - bX_{t-1}\psi_{t-1,b},\\
\psi_{t,\alpha} &= \frac{\partial e_t}{\partial \alpha} = -1 - bX_{t-1}\frac{\partial e_{t-1}}{\partial \alpha} = -1 - bX_{t-1}\psi_{t-1,\alpha},
\end{aligned}$$

with $\psi_{1,a} = \psi_{1,b} = \psi_{1,\alpha} = 0$. For example, the derivative with respect to b:

$$\psi_{2,b} = \tfrac{\partial e_2}{\partial b} = -(1.2)(0) - (0.3)(1.2)(0) = 0,$$

$$\psi_{3,b} = \tfrac{\partial e_3}{\partial b} = -(0.8)(-0.18) - (0.3)(0.8)(0) = 0.1440,$$
$$\psi_{4,b} = \tfrac{\partial e_4}{\partial b} = -(1.9)(1.1232) - (0.3)(1.9)(0.1440) = -2.1341 - 0.0821 = -2.2162,$$
$$\psi_{5,b} = \tfrac{\partial e_5}{\partial b} = -(0.4)(-1.5002) - (0.3)(0.4)(-2.2162) = 0.6001 + 0.2659 = 0.8660,$$
$$\psi_{6,b} = \tfrac{\partial e_6}{\partial b} = -(1.5)(1.0200) - (0.3)(1.5)(0.8660) = -1.5300 - 0.3897 = -1.9198.$$

Each row uses ψ from the previous row; for example $\psi_{4,b}$ uses $\psi_{3,b} = 0.1440$. If the recursive term were ignored, one would get $\psi_{6,b} = -1.5300$, which is wrong by about 20%.

Table 4: Errors and their derivatives at $\boldsymbol{\theta}^{(0)}$.

t	X_t	e_t	$\psi_{t,a} = \partial e_t/\partial a$	$\psi_{t,b} = \partial e_t/\partial b$	$\psi_{t,\alpha} = \partial e_t/\partial \alpha$
1	1.2	0	0	0	0
2	0.8	−0.1800	1.2000	0.0000	−1.0000
3	1.9	1.1232	0.5120	0.1440	−0.7600
4	0.4	−1.5002	1.6082	−2.2162	−0.5668
5	1.5	1.0200	0.2070	0.8660	−0.9320
6	1.1	−0.4590	1.4068	−1.9198	−0.5806

Check by finite differences. Recomputing e_6 with each parameter shifted by $\pm 10^{-6}$ gives $\psi_{6,a} = \partial e_6/\partial a \approx 1.406841$, $\psi_{6,b} = \partial e_6/\partial b \approx -1.919753$, $\psi_{6,\alpha} = \partial e_6/\partial \alpha \approx -0.580607$, identical to the results of the recursive formulas to six decimal places.

13.3 One iteration step

From the table, $Q = \sum_{t=2}^{6} e_t^2 = 4.7958$ and $\hat{\sigma}^2 = Q/5 = 0.9592$. The matrix $\mathbf{J}$ consists of the last three columns of the table ($\psi_{t,a}$, $\psi_{t,b}$, $\psi_{t,\alpha}$) for $t = 2, \ldots, 6$. Then

$$\mathbf{J}'\mathbf{J} = \begin{pmatrix} 6.3104 & -6.0117 & -3.5104 \\ -6.0117 & 9.3676 & 1.4542 \\ -3.5104 & 1.4542 & 3.1046 \end{pmatrix}, \qquad \mathbf{J}'\mathbf{e} = \begin{pmatrix} -2.4881 \\ 5.2510 \\ -0.5074 \end{pmatrix}.$$

One entry as an example: $(\mathbf{J}'\mathbf{e})_1 = \sum_t \psi_{t,a} e_t = (1.2)(-0.18) + (0.512)(1.1232) + (1.6082)(-1.5002) + (0.207)(1.02) + (1.4068)(-0.459) = -2.4881$. The gradient and approximate Hessian are $\mathbf{g} = 2\mathbf{J}'\mathbf{e}$ and $\mathbf{H} = 2\mathbf{J}'\mathbf{J}$. Solving $(\mathbf{J}'\mathbf{J})\boldsymbol{\delta} = \mathbf{J}'\mathbf{e}$ gives $\boldsymbol{\delta} = (-2.0812,\ -0.4145,\ -2.3226)'$, so

$$\boldsymbol{\theta}^{(1)} = \boldsymbol{\theta}^{(0)} - \boldsymbol{\delta} = (1.6812,\ 0.7145,\ 2.8226)', \qquad Q(\boldsymbol{\theta}^{(1)}) = 3.1364 < 4.7958.$$

With only five errors, the step is very large; this example is meant for checking the computations, not for obtaining meaningful estimates.

14 A manual Gauss–Newton iteration example

This section runs Algorithm 1 by hand for two iterations, with all intermediate computations. So that the computations can be followed by hand, a model with two parameters is used.

14.1 Data and model

The following eleven observations were generated from $X_t = 0.5X_{t-1} + 0.4X_{t-1}e_{t-1} + e_t$ with $e_t \sim N(0, 1)$ and rounded to one decimal place:

t	1	2	3	4	5	6	7	8	9	10	11
X_t	-0.6	-0.2	-1.2	-0.5	0.0	0.1	1.6	1.9	0.8	-0.5	0.4

The model to be estimated (without a constant), in the convention of this document:

$$X_t + aX_{t-1} = b\,X_{t-1}e_{t-1} + e_t, \qquad \boldsymbol{\theta} = (a, b)',$$

so the true values are $a = -0.5$ and $b = 0.4$. The largest lag is $\gamma = 1$, so $e_1 = 0$ and $n = N - \gamma = 10$. The formulas used, from (3) and (10):

$$e_t = X_t + aX_{t-1} - bX_{t-1}e_{t-1},$$
$$\psi_{t,a} = \frac{\partial e_t}{\partial a} = X_{t-1} - bX_{t-1}\psi_{t-1,a}, \qquad \psi_{t,b} = \frac{\partial e_t}{\partial b} = -X_{t-1}e_{t-1} - bX_{t-1}\psi_{t-1,b},$$

with $e_1 = \psi_{1,a} = \psi_{1,b} = 0$.

14.2 Step 0: starting values

Start from the AR(1) model with $b = 0$: $X_t = -aX_{t-1} + e_t$. The OLS estimator (regression without a constant) is

$$a^{(0)} = -\frac{\sum_{t=2}^{11} X_{t-1}X_t}{\sum_{t=2}^{11} X_{t-1}^2} = -\frac{5.08}{9.16} = -0.5546, \qquad b^{(0)} = 0, \qquad \boldsymbol{\theta}^{(0)} = (-0.5546,\ 0)'.$$

14.3 First iteration ($s = 0$)

Errors and derivatives. Since $b^{(0)} = 0$, the recursion simplifies: $e_t = X_t + aX_{t-1}$, $\psi_{t,a} = X_{t-1}$, and $\psi_{t,b} = -X_{t-1}e_{t-1}$. For example

$$e_2 = -0.2 + (-0.5546)(-0.6) = 0.1328, \qquad \psi_{2,b} = -X_1 e_1 = -(-0.6)(0) = 0,$$
$$e_3 = -1.2 + (-0.5546)(-0.2) = -1.0891, \qquad \psi_{3,b} = -X_2 e_2 = -(-0.2)(0.1328) = 0.0266,$$
$$e_4 = -0.5 + (-0.5546)(-1.2) = 0.1655, \qquad \psi_{4,b} = -X_3 e_3 = -(-1.2)(-1.0891) = -1.3069.$$

All rows are summarised in Table 5. The columns on the right are the products needed for Q, $\mathbf{J}'\mathbf{J}$, and $\mathbf{J}'\mathbf{e}$.

Table 5: First iteration, computed at $\boldsymbol{\theta}^{(0)} = (-0.5546,\ 0)'$.

t	X_t	e_t	$\psi_{t,a}$	$\psi_{t,b}$	e_t^2	$\psi_{t,a}^2$	$\psi_{t,a}\psi_{t,b}$	$\psi_{t,b}^2$	$\psi_{t,a}e_t$	$\psi_{t,b}e_t$
2	-0.2000	0.1328	-0.6000	0.0000	0.0176	0.3600	0.0000	0.0000	-0.0797	0.0000
3	-1.2000	-1.0891	-0.2000	0.0266	1.1861	0.0400	-0.0053	0.0007	0.2178	-0.0289
4	-0.5000	0.1655	-1.2000	-1.3069	0.0274	1.4400	1.5683	1.7080	-0.1986	-0.2163
5	0.0000	0.2773	-0.5000	0.0828	0.0769	0.2500	-0.0414	0.0068	-0.1386	0.0229
6	0.1000	0.1000	0.0000	0.0000	0.0100	0.0000	0.0000	0.0000	0.0000	0.0000
7	1.6000	1.5445	0.1000	-0.0100	2.3856	0.0100	-0.0010	0.0001	0.1545	-0.0154
8	1.9000	1.0127	1.6000	-2.4713	1.0255	2.5600	-3.9540	6.1072	1.6203	-2.5026
9	0.8000	-0.2537	1.9000	-1.9241	0.0644	3.6100	-3.6557	3.7020	-0.4821	0.4882
10	-0.5000	-0.9437	0.8000	0.2030	0.8905	0.6400	0.1624	0.0412	-0.7549	-0.1915
11	0.4000	0.6773	-0.5000	-0.4718	0.4587	0.2500	0.2359	0.2226	-0.3386	-0.3196
Sum					6.1427	9.1600	-5.6909	11.7886	0.0000	-2.7632

Forming $\mathbf{J}'\mathbf{J}$ and $\mathbf{J}'\mathbf{e}$. From the row of sums:

$$
\begin{aligned}
Q^{(0)} &= \sum e_t^2 = 6.1427, \\
\mathbf{J}'\mathbf{J} &= \begin{pmatrix} \sum \psi_{t,a}^2 & \sum \psi_{t,a}\psi_{t,b} \\ \sum \psi_{t,a}\psi_{t,b} & \sum \psi_{t,b}^2 \end{pmatrix} = \begin{pmatrix} 9.1600 & -5.6909 \\ -5.6909 & 11.7886 \end{pmatrix}, \\
\mathbf{J}'\mathbf{e} &= \begin{pmatrix} \sum \psi_{t,a} e_t \\ \sum \psi_{t,b} e_t \end{pmatrix} = \begin{pmatrix} 0.0000 \\ -2.7632 \end{pmatrix}.
\end{aligned}
$$

The gradient component for a is zero because $a^{(0)}$ is the OLS estimator when $b = 0$: at that point Q is already minimised with respect to a. The component for b is not zero, so Q can still be decreased by changing b.

Inverse of a 2×2 matrix. For $\mathbf{M} = \begin{pmatrix} p & r \\ r & w \end{pmatrix}$, $\mathbf{M}^{-1} = \dfrac{1}{pw - r^2} \begin{pmatrix} w & -r \\ -r & p \end{pmatrix}$. Here

$$
\det(\mathbf{J}'\mathbf{J}) = (9.1600)(11.7886) - (-5.6909)^2 = 107.9836 - 32.3863 = 75.5973,
$$

$$
(\mathbf{J}'\mathbf{J})^{-1} = \frac{1}{75.5973} \begin{pmatrix} 11.7886 & 5.6909 \\ 5.6909 & 9.1600 \end{pmatrix} = \begin{pmatrix} 0.15594 & 0.07528 \\ 0.07528 & 0.12117 \end{pmatrix}.
$$

Correction vector and new value.

$$
\boldsymbol{\delta}^{(0)} = (\mathbf{J}'\mathbf{J})^{-1}\mathbf{J}'\mathbf{e} = \begin{pmatrix} (0.15594)(0.0000) + (0.07528)(-2.7632) \\ (0.07528)(0.0000) + (0.12117)(-2.7632) \end{pmatrix} = \begin{pmatrix} -0.2080 \\ -0.3348 \end{pmatrix},
$$

$$
\boldsymbol{\theta}^{(1)} = \boldsymbol{\theta}^{(0)} - \boldsymbol{\delta}^{(0)} = \begin{pmatrix} -0.5546 - (-0.2080) \\ 0 - (-0.3348) \end{pmatrix} = \begin{pmatrix} -0.3466 \\ 0.3348 \end{pmatrix}.
$$

14.4 Second iteration ($s = 1$)

Errors and derivatives. Now $b^{(1)} = 0.3348 \neq 0$, so the recursive terms come into play. For example:

$$
\begin{aligned}
e_2 &= -0.2 + (-0.3466)(-0.6) - (0.3348)(-0.6)(0) = 0.0079, \\
e_3 &= -1.2 + (-0.3466)(-0.2) - (0.3348)(-0.2)(0.0079) = -1.1302, \\
e_4 &= -0.5 + (-0.3466)(-1.2) - (0.3348)(-1.2)(-1.1302) = -0.5382, \\
\psi_{3,a} &= X_2 - bX_2\psi_{2,a} = -0.2 - (0.3348)(-0.2)(-0.6) = -0.2402, \\
\psi_{4,a} &= X_3 - bX_3\psi_{3,a} = -1.2 - (0.3348)(-1.2)(-0.2402) = -1.2965, \\
\psi_{3,b} &= -X_2 e_2 - bX_2\psi_{2,b} = -(-0.2)(0.0079) - (0.3348)(-0.2)(0) = 0.0016, \\
\psi_{4,b} &= -X_3 e_3 - bX_3\psi_{3,b} = -(-1.2)(-1.1302) - (0.3348)(-1.2)(0.0016) = -1.3555.
\end{aligned}
$$

Checking the decrease in Q. $Q^{(1)} = 5.0500 < Q^{(0)} = 6.1427$, so the full step ($\lambda = 1$) is accepted without step-halving.

Table 6: Second iteration, computed at $\boldsymbol{\theta}^{(1)} = (-0.3466,\ 0.3348)'$.

t	X_t	e_t	$\psi_{t,a}$	$\psi_{t,b}$	e_t^2	$\psi_{t,a}^2$	$\psi_{t,a}\psi_{t,b}$	$\psi_{t,b}^2$	$\psi_{t,a}e_t$	$\psi_{t,b}e_t$
2	−0.2000	0.0079	−0.6000	0.0000	0.0001	0.3600	0.0000	0.0000	−0.0048	0.0000
3	−1.2000	−1.1302	−0.2402	0.0016	1.2772	0.0577	−0.0004	0.0000	0.2714	−0.0018
4	−0.5000	−0.5382	−1.2965	−1.3555	0.2896	1.6809	1.7575	1.8375	0.6977	0.7295
5	0.0000	0.0832	−0.7170	−0.4960	0.0069	0.5141	0.3557	0.2460	−0.0597	−0.0413
6	0.1000	0.1000	0.0000	0.0000	0.0100	0.0000	0.0000	0.0000	0.0000	0.0000
7	1.6000	1.5620	0.1000	−0.0100	2.4398	0.0100	−0.0010	0.0001	0.1562	−0.0156
8	1.9000	0.5087	1.5464	−2.4938	0.2588	2.3914	−3.8565	6.2192	0.7867	−1.2687
9	0.8000	−0.1821	0.9163	0.6199	0.0332	0.8395	0.5680	0.3842	−0.1669	−0.1129
10	−0.5000	−0.7285	0.5546	−0.0203	0.5307	0.3076	−0.0113	0.0004	−0.4040	0.0148
11	0.4000	0.4513	−0.4072	−0.3676	0.2037	0.1658	0.1497	0.1352	−0.1838	−0.1659
Sum					5.0500	6.3270	−1.0384	8.8226	1.0930	−0.8618

Forming $\mathbf{J}'\mathbf{J}$ and $\mathbf{J}'\mathbf{e}$.

$$\mathbf{J}'\mathbf{J} = \begin{pmatrix} 6.3270 & -1.0384 \\ -1.0384 & 8.8226 \end{pmatrix}, \qquad \mathbf{J}'\mathbf{e} = \begin{pmatrix} 1.0930 \\ -0.8618 \end{pmatrix},$$

$$\det(\mathbf{J}'\mathbf{J}) = (6.3270)(8.8226) - (-1.0384)^2 = 55.8206 - 1.0783 = 54.7423,$$

$$(\mathbf{J}'\mathbf{J})^{-1} = \frac{1}{54.7423}\begin{pmatrix} 8.8226 & 1.0384 \\ 1.0384 & 6.3270 \end{pmatrix} = \begin{pmatrix} 0.16117 & 0.01897 \\ 0.01897 & 0.11558 \end{pmatrix}.$$

Correction vector and new value.

$$\boldsymbol{\delta}^{(1)} = \begin{pmatrix} (0.16117)(1.0930) + (0.01897)(-0.8618) \\ (0.01897)(1.0930) + (0.11558)(-0.8618) \end{pmatrix} = \begin{pmatrix} 0.17616 - 0.01635 \\ 0.02073 - 0.09961 \end{pmatrix} = \begin{pmatrix} 0.1598 \\ -0.0789 \end{pmatrix},$$

$$\boldsymbol{\theta}^{(2)} = \boldsymbol{\theta}^{(1)} - \boldsymbol{\delta}^{(1)} = \begin{pmatrix} -0.3466 - 0.1598 \\ 0.3348 - (-0.0789) \end{pmatrix} = \begin{pmatrix} -0.5064 \\ 0.4137 \end{pmatrix}, \qquad Q^{(2)} = 4.7888.$$

14.5 Further iterations and convergence

Continuing the same procedure gives:

s	$a^{(s)}$	$b^{(s)}$	$Q^{(s)}$	relative change in Q
0	−0.5546	0.0000	6.14270	–
1	−0.3466	0.3348	5.05000	1.8×10^{-1}
2	−0.5064	0.4137	4.78883	5.2×10^{-2}
3	−0.5239	0.4126	4.78699	3.8×10^{-4}
4	−0.5237	0.4137	4.78698	2×10^{-6}
5	−0.5238	0.4135	4.78698	$< 10^{-8}$

The iteration stops once the relative change in Q is below 10^{-8}. The final estimates are $\hat{a} = -0.5238$, $\hat{b} = 0.4136$, and $\hat{\sigma}^2 = Q/n = 4.78698/10 = 0.4787$, close to the true values $a = -0.5$ and $b = 0.4$. Note the general pattern: Q falls quickly in the first few iterations, then its changes shrink rapidly once the estimate is close to the minimum.

Note

The numbers in the tables are rounded to four decimal places. If the computations are repeated with the rounded numbers, the last digit may differ slightly. With only 10 errors these estimates are of course very imprecise; the example is meant to show the mechanics of the iteration.

15 Simulation study

Data were generated from $X_t - 0.4X_{t-1} = 0.5 + 0.3X_{t-1}e_{t-1} + e_t$, i.e. $a = -0.4$, $b = 0.3$, $\alpha = 0.5$, with $\sigma^2 = 1$ and $N = 500$ (the first 100 values were discarded so that the effect of the initial values disappears). Table 7 shows the Gauss–Newton iterations (full step) for the first replication, starting from AR(1) with $b = 0$.

Table 7: Gauss–Newton iterations on one simulated data set.

Iteration	a	b	α	$\hat{\sigma}^2$
0	−0.6093	0.0000	0.4976	1.22797
1	−0.4070	0.1683	0.5463	1.01386
2	−0.4197	0.2690	0.4610	0.94040
3	−0.4111	0.2984	0.4624	0.93622
4	−0.4003	0.2929	0.4775	0.93551
5	−0.4003	0.2945	0.4765	0.93549
8	−0.3999	0.2942	0.4771	0.93549

Standard errors (19) at the final values: 0.038, 0.015, 0.056.

Table 8: Monte Carlo with 200 replications ($N = 500$). "SE" is the average of the standard errors (19).

	$a = -0.4$	$b = 0.3$	$\alpha = 0.5$
One starting value (AR, $b = 0$)			
Mean of estimates	−0.408	0.283	0.510
Standard deviation of estimates	0.053	0.052	0.056
Coverage of the 95% confidence interval	0.875	0.795	0.960
Estimate of b off by > 0.1	21 of 200 (19 of them local minima)		
Multi-start, $b^{(0)} \in \{-0.4, -0.2, 0, 0.2, 0.4\}$			
Mean of estimates	−0.397	0.300	0.509
Standard deviation of estimates	0.040	0.020	0.058
Mean SE	0.038	0.010	0.058
Coverage of the 95% confidence interval	0.955	0.915	0.955
Estimate of b off by > 0.1		2 of 200	

Two lessons: (1) even for the simplest model with $N = 500$, about 10% of the estimates from a single starting value get trapped in a local minimum ("local minimum" here means that an iteration started from the true values reaches a smaller Q); multi-start solves this. (2) The standard errors (19) for b tend to be too small (coverage 91.5%).

16 Verification with real data

Subba Rao & Gabr (1984) report bilinear models for three series whose data they also list: annual sunspot numbers 1700–1955, annual Canadian lynx trappings 1821–1934, and monthly West German unemployment 1948–1980. The method in this document was implemented independently and run on the same data; the results are compared with those they report.

16.1 Model S: sunspots

Data 1700–1920 ($N = 221$), $\gamma = 10$, $n = 211$. AR lags $\mathcal{K} = \{1, 2, 9\}$ and seven bilinear terms:

$$X_t + a_1 X_{t-1} + a_2 X_{t-2} + a_9 X_{t-9} = \alpha + \sum_{(i,j)\in\mathcal{B}} b_{ij} X_{t-i} e_{t-j} + e_t,$$

with $\mathcal{B} = \{(2,1), (8,1), (1,3), (4,3), (1,6), (2,4), (3,2)\}$. Gauss–Newton from the AR model with all $b = 0$ converges to:

Table 9: Model S: reported values and recomputed results.

Parameter	Reported	Recomputed	SE (19)	t
a_1	-1.5012	-1.5012	0.0439	-34.2
a_2	0.7670	0.7670	0.0452	17.0
a_9	-0.1152	-0.1152	0.0211	-5.5
$b_{2,1}$	$-0.1458^\dagger$	-0.014575	0.0010	-14.3
$b_{8,1}$	0.006312	0.006312	0.0010	6.6
$b_{1,3}$	-0.007152	-0.007152	0.0009	-7.9
$b_{4,3}$	$-006047^\dagger$	0.006047	0.0012	4.9
$b_{1,6}$	0.003619	0.003619	0.0007	4.9
$b_{2,4}$	0.004334	0.004333	0.0012	3.7
$b_{3,2}$	0.001782	0.001781	0.0015	1.2
α	6.886	6.886	1.48	4.6
$\hat\sigma^2$ / NAIC	124.33 / 4.927	124.33 / 4.927		

† misprint; see Section 17.

16.2 Model L1: log lynx

$\log_{10}$ data 1821–1920 ($N = 100$), $\gamma = 12$, $n = 88$, AR lags $\{1, 2, 3, 4, 9, 12\}$, and bilinear terms $(3, 9)$, $(9, 9)$, $(6, 2)$, $(1, 1)$, $(2, 7)$, $(4, 2)$. Adding the terms one at a time in that order gives $\hat\sigma^2 = 0.02227$ and NAIC $= -3.509$ (reported: 0.0223 and -3.508). The recomputed coefficients, all within ± 0.005 of those reported:

$$\hat a = (-0.771,\ 0.090,\ -0.082,\ 0.262,\ -0.225,\ 0.246),$$
$$\hat b = (-0.790,\ 0.480,\ 0.391,\ 0.134,\ 0.080,\ -0.322), \qquad \hat\alpha = 1.492.$$

16.3 Model L2: raw lynx

Raw data 1821–1920, $\gamma = 10$, $n = 90$, AR lags $\{1, 2, 7, 8\}$, and seven bilinear terms. The data were divided by 100 before estimation for numerical stability (see Section 18), and the coefficients were then transformed back to the original scale. With the terms added in the order

$(1,1),(3,7),(7,6),(3,5),(4,1),(9,10),(8,10)$, the result is

$$X_t - 1.2601X_{t-1} + 0.6559X_{t-2} + 0.1572X_{t-7} - 0.3375X_{t-8} = 343.6 + \sum b_{ij}X_{t-i}e_{t-j} + e_t,$$

with $\hat{\sigma}^2 = 477\,946$ (reported 477 964.68), and all seven coefficients b agree to two significant figures, for example $b_{8,10} = -4.15 \times 10^{-4}$ (reported -4.151×10^{-4}). If the order of addition is reversed, the estimation ends at a different minimum ($\hat{\sigma}^2 = 486\,637$) that is not invertible.

16.4 Not yet verified

The model for the unemployment data was not recomputed. Those data need to be transformed by ordinary and 12-month seasonal differencing, and the model contains eleven bilinear terms with coefficients of order 10^{-5}–10^{-6}.

17 Notes on figures printed in the book

While recomputing, several figures were found to be misprinted or confusing in Subba Rao & Gabr (1984). The models referred to are Models S, L1, and L2 of Section 16.

Table 10: Printed figures and correct values.

Location	Printed	Correct value and evidence
Model S	$b_{2,1} = -0.1458$	-0.01458; with the printed value the error recursion explodes.
Model S	$b_{4,3} = -006047$	$+0.006047$; $\hat{\sigma}^2 = 124.33$ is reached only with the positive sign.
Subset AR, sunspots	$a_9 = -0.1450$	-0.1500 from OLS; $\hat{\sigma}^2 = 203.21$ identical to the reported value.
Model L1	$\alpha = -1.48629$	$+1.486$; consistent with $\alpha \approx \bar{X}(1 + \sum a_k)$ (Section 7).
Model L1	$-b_{4,2}X_{t-4}e_{t-2}$ with $b_{4,2} = -0.3212$	the minus sign is printed twice; the effective coefficient is -0.3212.
Model L2	"$X(t) = 1.2598X(t-1) + \ldots$"	in the convention of this document: $X_t - 1.2598X_{t-1} + 0.6548X_{t-2} + 0.1564X_{t-7} - 0.3392X_{t-8} - 336.2 = \ldots$
Lynx data	1867: 4245; 1871: 353	R's built-in `lynx` dataset uses 4254 and 358; both versions give almost the same results.
Normality test	$\mathrm{Var}(G_1) = 6\sum\rho^2$	$\frac{6}{N}\sum\rho^2$; likewise for G_2 (Section 11).

18 Numerical pitfalls and recommendations

1. **Local minima.** $Q(\boldsymbol{\theta})$ is not convex. Use staged addition of terms and multi-start for b.
2. **Order dependence.** For Model L2, the order in which terms are added determines the minimum reached. Compare the results of several orders and choose the smallest invertible Q.
3. **Invertibility.** Check every candidate model with (20). Without this check, automatic selection tends to pick non-invertible models with a spuriously small $\hat{\sigma}^2$ (Figure 3).

4. **Data scale.** If X_t takes values in the hundreds or thousands, the coefficients b become very small and the matrix $\mathbf{J}'\mathbf{J}$ is numerically hard to invert. Scale the data as $\tilde{X}_t = X_t/c$. Since the errors are scaled too, $\tilde{e}_t = e_t/c$, and the model equation becomes $\tilde{X}_t + \sum a_k \tilde{X}_{t-k} = \alpha/c + \sum (b_{ij}c)\tilde{X}_{t-i}\tilde{e}_{t-j} + \tilde{e}_t$. So a_k is unchanged, $\alpha = c\,\tilde{\alpha}$, $b_{ij} = \tilde{b}_{ij}/c$, and $\sigma^2 = c^2\tilde{\sigma}^2$.

5. **Step-size control.** Use step-halving or Levenberg–Marquardt, and flag a failure when Q is infinite.

6. **Model selection stopping rule.** Set a strict rule, for example stop when AIC increases.

7. **Inference.** Report the standard errors (19) as approximations; use the bootstrap when formal conclusions are needed.

8. **Normality test.** Use $\mathrm{Var}(G_1) = \frac{6}{N}\sum \rho^2$.

Summary of what the estimation function needs to contain

1. The error recursion (3) and $Q(\boldsymbol{\theta})$.
2. The recursive derivatives (10), including the MA extension when needed.
3. The Gauss–Newton iteration (17) with step-size control (Algorithm 1).
4. Starting values from OLS of the AR model; multi-start for b.
5. AIC/NAIC (18) with the same $n = N - \gamma$ for all models.
6. Subset AR selection (exhaustive search) and stepwise addition of bilinear terms.
7. Standard errors (19) and the invertibility diagnostic (20).
8. Residual tests W, u_1, u_2 (with $1/N$).
9. One-step forecasts (recursion) and h-step forecasts (conditional expectation or Monte Carlo).
10. Testing the function: it must reproduce Models S, L1, and L2.

References

- Grahn, T. (1995). A conditional least squares approach to bilinear time series estimation. *J. Time Series Anal.* 16, 509–529.
- Granger, C. W. J. & Andersen, A. P. (1978). On the invertibility of time series models. *Stoch. Proc. Appl.* 8, 83–92.
- Ling, S., Peng, L. & Zhu, F. Inference for a special bilinear time series model. arXiv:1405.3029.
- Lomnicki, Z. A. (1961). Tests for departure from normality in the case of linear stochastic processes. *Metrika* 4, 37–62.
- Marquardt, D. W. (1963). An algorithm for least squares estimation of nonlinear parameters. *J. SIAM* 11, 431–441.
- Quinn, B. G. (1982). Stationarity and invertibility of simple bilinear models. *Stoch. Proc. Appl.* 12, 225–230.
- Subba Rao, T. (1981). On the theory of bilinear time series models. *J. R. Statist. Soc. B* 43, 244–255.
- Subba Rao, T. & Gabr, M. M. (1984). *An Introduction to Bispectral Analysis and Bilinear Time Series Models.* Lecture Notes in Statistics 24, Springer.